Theory of Matrices

(MMT126J)

Lecture Notes for Undergraduate Students

$$\det(A-\lambda I)=0 \bullet \operatorname{Tr}(A)=\sum_{i=1}^{n} a_{ii} \bullet CHT: p_A(A)=0 \bullet |\langle u,v\rangle| \le \|u\|\|v\|$$

Prepared and Written by

Dr. Aijaz

Assistant Professor of Mathematics

JK Higher Education Department

Study Materials for UG Mathematics, PG Mathematics and Entrance Examinations

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CHAPTER

1

MATRIX ALGEBRA AND STRUCTURAL PROPERTIES

Matrices provide a convenient way of arranging numbers and performing calculations involving several quantities at once. They are widely used in mathematics, science, engineering, economics, and computer applications. In particular, matrices play an important role in solving systems of linear equations and studying linear transformations.

In this chapter, we introduce matrices with real and complex entries and study their basic algebraic operations. Special attention is given to matrix multiplication, which differs from ordinary multiplication of numbers because the order of multiplication generally matters. We shall also study the transpose and inverse of a matrix, their important properties, and the reversal laws associated with these operations. The remaining sections deal with conjugate transpose, special classes of matrices, matrix decompositions, and the adjoint of a square matrix.

1.1 Algebraic Operations on Matrices over Real and Complex Fields

Definition 1.1.1 (Matrix)**.** A **matrix** is a rectangular arrangement of numbers or other mathematical quantities in rows and columns, usually enclosed within square brackets.

A matrix having m rows and n columns is called a matrix of **order** $m \times n$ and is written as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = (a_{ij})_{m\times n}.$$

The numbers a_{ij} are called the **entries** or **elements** of the matrix, where a_{ij} denotes the entry in the i-th row and j-th column.

Example 1.1.2. Consider the matrix

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \end{bmatrix}.$$

Since A has two rows and three columns, its order is 2×3. Here $a_{12} = -1$, $a_{23} = 5$, and $a_{21} = 0$.

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The set of all $m \times n$ matrices with real entries is denoted by $M_{m\times n}(\mathbb{R})$, while $M_{m\times n}(\mathbb{C})$ denotes the set of all $m \times n$ matrices with complex entries. For square matrices of order $n \times n$, we write $M_n(\mathbb{R})$ or $M_n(\mathbb{C})$.

Throughout this chapter, $\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$, unless stated otherwise.

1.1.1 Some Basic Types of Matrices

Before discussing matrix operations, we recall a few important types of matrices.

(i) **Row matrix:** A matrix having only one row, such as $\begin{bmatrix} 2 & -1 & 4 \end{bmatrix}$.

(ii) **Column matrix:** A matrix having only one column, such as $\begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$.

(iii) **Square matrix:** A matrix having an equal number of rows and columns, such as $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

(iv) **Zero matrix:** A matrix whose entries are all zero. It is denoted by 0 or O.

(v) **Diagonal matrix:** A square matrix whose entries outside the main diagonal are zero, such as $\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$.

(vi) **Scalar matrix:** A diagonal matrix whose diagonal entries are all equal, such as $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

(vii) **Identity matrix:** A diagonal matrix whose diagonal entries are all 1. The identity matrix of order n is denoted by I_n.

For example,

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Two matrices are said to be **equal** if they have the same order and their corresponding entries are equal.

1.1.2 Matrix Addition and Scalar Multiplication

Definition 1.1.3 (Matrix Addition)**.** Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same order $m \times n$. Their sum is obtained by adding corresponding entries:

$$A + B = (a_{ij} + b_{ij})_{m\times n}.$$

Thus, $(A + B)_{ij} = a_{ij} + b_{ij}$. Two matrices can be added only when they have the same order.

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Example 1.1.4. Let $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$. Then

$$A + B = \begin{bmatrix} 2+1 & -1+5 \\ 3-2 & 4+3 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 7 \end{bmatrix}.$$

Definition 1.1.5 (Scalar Multiplication)**.** Let $A = (a_{ij})$ be an $m \times n$ matrix and let $\alpha \in \mathbb{K}$. The scalar multiple αA is obtained by multiplying every entry of A by α:

$$\alpha A = (\alpha a_{ij})_{m \times n}.$$

Example 1.1.6. If $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, then

$$3A = \begin{bmatrix} 3 & -6 \\ 9 & 0 \end{bmatrix}, \qquad -A = \begin{bmatrix} -1 & 2 \\ -3 & 0 \end{bmatrix}.$$

Matrix subtraction is defined by $A - B = A + (-B)$. Hence, corresponding entries of B are subtracted from those of A.

Theorem 1.1.7 (Properties of Addition and Scalar Multiplication)**.** *Let $A, B, C \in M_{m \times n}(\mathbb{K})$ and $\alpha, \beta \in \mathbb{K}$. Then the following properties hold:*

(i) $A + B = B + A$ *(commutative law).*

(ii) $(A + B) + C = A + (B + C)$ *(associative law).*

(iii) $A + 0 = A$ *(additive identity).*

(iv) $A + (-A) = 0$ *(additive inverse).*

(v) $\alpha(A + B) = \alpha A + \alpha B$.

(vi) $(\alpha + \beta)A = \alpha A + \beta A$.

(vii) $\alpha(\beta A) = (\alpha\beta)A$.

(viii) $1A = A$ *and* $0A = 0$.

Proof. All these properties follow directly from the corresponding properties of real or complex numbers, since addition and scalar multiplication are performed entry by entry.

For instance, if $A = (a_{ij})$ and $B = (b_{ij})$, then

$$(A + B)_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (B + A)_{ij}.$$

Thus, $A + B = B + A$. Similarly,

$$\big(\alpha(A + B)\big)_{ij} = \alpha(a_{ij} + b_{ij}) = \alpha a_{ij} + \alpha b_{ij} = (\alpha A + \alpha B)_{ij}.$$

Hence, $\alpha(A + B) = \alpha A + \alpha B$. The remaining properties are verified in the same way. ∎

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1.1.3 Matrix Multiplication

Unlike addition, matrix multiplication is not performed by multiplying corresponding entries. Instead, each entry of the product is obtained by multiplying the entries of a row of the first matrix by the corresponding entries of a column of the second matrix and adding the resulting products.

Definition 1.1.8 (Matrix Multiplication)**.** Let $A = (a_{ij})$ be an $m \times p$ matrix and $B = (b_{jk})$ be a $p \times n$ matrix. Their product AB is the $m \times n$ matrix whose (i,k)-th entry is

$$(AB)_{ik} = \sum_{j=1}^{p} a_{ij}b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{ip}b_{pk}.$$

Thus, the entry in the i-th row and k-th column of AB is obtained by multiplying the i-th row of A with the k-th column of B.

For the product AB to exist, the number of columns of A must equal the number of rows of B. The order of the resulting matrix is determined by the remaining two dimensions:

$$\underset{m\times p}{A}\ \underset{p\times n}{B} = \underset{m\times n}{AB}.$$

In particular, if A is of order 2×3 and B is of order 3×4, then AB exists and has order 2×4.

Example 1.1.9. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 0 \\ 1 & 4 \end{bmatrix}.$$

To obtain the first entry of AB, multiply the first row of A by the first column of B: $1(2) + 2(1) = 4$. Continuing in the same way,

$$AB = \begin{bmatrix} 1(2)+2(1) & 1(0)+2(4) \\ 3(2)-1(1) & 3(0)-1(4) \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 5 & -4 \end{bmatrix}.$$

Example 1.1.10. Find the product AB, where

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ -1 & 3 \end{bmatrix}.$$

Solution. Since A is of order 2×3 and B is of order 3×2, the product AB exists and has order 2×2. By multiplying rows with columns, we obtain

$$AB = \begin{bmatrix} 1(2)+2(1)+(-1)(-1) & 1(1)+2(0)+(-1)(3) \\ 0(2)+3(1)+2(-1) & 0(1)+3(0)+2(3) \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 1 & 6 \end{bmatrix}.$$

■

Theorem 1.1.11 (Properties of Matrix Multiplication)**.** *Let A, B, C be matrices over $\mathbb{K}$ of suitable orders, and let $\alpha \in \mathbb{K}$. Then:*

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(i) $(AB)C = A(BC)$ *(associative law).*

(ii) $A(B+C) = AB + AC$ *(left distributive law).*

(iii) $(A+B)C = AC + BC$ *(right distributive law).*

(iv) $\alpha(AB) = (\alpha A)B = A(\alpha B)$.

(v) $AI_n = I_m A = A$ *whenever* A *is of order* $m \times n$.

(vi) $A0 = 0$ *and* $0A = 0$, *whenever the products are defined.*

Proof. We illustrate the distributive law by comparing corresponding entries. Let $A = (a_{ij})$ be of order $m \times p$ and let $B = (b_{jk})$ and $C = (c_{jk})$ be of order $p \times n$. Then

$$\begin{aligned}\big(A(B+C)\big)_{ik} &= \sum_{j=1}^{p} a_{ij}(b_{jk} + c_{jk}) \\ &= \sum_{j=1}^{p} a_{ij}b_{jk} + \sum_{j=1}^{p} a_{ij}c_{jk} \\ &= (AB)_{ik} + (AC)_{ik}.\end{aligned}$$

Therefore, $A(B+C) = AB + AC$.

Similarly, for the associative law, let A, B, C have orders $m \times p$, $p \times q$, and $q \times n$, respectively. The (i,k)-th entries of $(AB)C$ and $A(BC)$ are both equal to

$$\sum_{j=1}^{p}\sum_{r=1}^{q} a_{ij}b_{jr}c_{rk}.$$

Hence, $(AB)C = A(BC)$. The other properties follow directly from the definition of matrix multiplication. ■

1.1.4 Non-Commutativity of Matrix Multiplication

For real or complex numbers, multiplication is commutative: $ab = ba$. However, matrix multiplication generally does not satisfy this property. Even when both AB and BA exist, they need not be equal.

Example 1.1.12. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}, \qquad BA = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}.$$

Thus, $AB \neq BA$. This shows that matrix multiplication is not commutative in general.

There are three possibilities when comparing AB and BA:

(i) One product may exist while the other does not.

(ii) Both products may exist but have different orders.

(iii) Both products may have the same order but different entries.

For example, if A has order 2×3 and B has order 3×2, then AB has order 2×2, while BA has order 3×3. Consequently, they cannot be equal.

Definition 1.1.13 (Commuting Matrices)**.** Two square matrices A and B of the same order are said to **commute** if $AB = BA$.

Example 1.1.14. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$. Then

$$AB = BA = \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix}.$$

Thus, A and B commute.

1.1.5 Consequences of Non-Commutativity

The non-commutativity of matrix multiplication leads to several important differences between ordinary algebra and matrix algebra.

Expansion of $(A+B)^2$. For real numbers, $(a+b)^2 = a^2 + 2ab + b^2$. For square matrices, however,

$$(A+B)^2 = A^2 + AB + BA + B^2.$$

The two middle terms cannot generally be combined because AB and BA need not be equal. Therefore, the familiar formula $(A+B)^2 = A^2 + 2AB + B^2$ is valid for matrices precisely when $AB = BA$.

Similarly,

$$(A-B)(A+B) = A^2 + AB - BA - B^2.$$

Hence, $(A-B)(A+B) = A^2 - B^2$ whenever A and B commute.

Example 1.1.15. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Since $AB \neq BA$, we cannot use the ordinary binomial formula. Instead, we must write $(A+B)^2 = A^2 + AB + BA + B^2$.

Product of Two Non-Zero Matrices. For real numbers, $ab = 0$ implies $a = 0$ or $b = 0$. This property does not hold for matrices.

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Example 1.1.16. Consider

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}.$$

Although $A \neq 0$ and $B \neq 0$, we have

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0.$$

Thus, the product of two non-zero matrices can be the zero matrix.

Failure of the Cancellation Law. For non-zero real numbers, $ab = ac$ implies $b = c$. For matrices, however, $AB = AC$ need not imply $B = C$, even when $A \neq 0$.

To see this, take A and B from the preceding example and let $C = 0$. Then $AB = AC = 0$, but $B \neq C$. Hence, cancellation is not valid in general.

We shall see later that cancellation is valid whenever the matrix being cancelled is invertible.

1.1.6 Special Cases Where Matrices Commute

Although matrix multiplication is generally non-commutative, the following are important cases in which two square matrices of the same order commute.

(i) **Identity matrix:** Every square matrix A commutes with I_n, since $AI_n = I_nA = A$.

(ii) **Scalar matrices:** For every scalar α, we have $A(\alpha I_n) = (\alpha I_n)A = \alpha A$.

(iii) **Diagonal matrices:** Any two diagonal matrices of the same order commute.

(iv) **Powers of the same matrix:** For non-negative integers r, s, we have $A^rA^s = A^{r+s} = A^sA^r$, where $A^0 = I_n$.

(v) **Matrix and its inverse:** If A is invertible, then $AA^{-1} = A^{-1}A = I_n$.

1.2 Transpose of a Matrix and Its Properties

Definition 1.2.1 (Transpose of a Matrix)**.** Let $A = (a_{ij})$ be an $m \times n$ matrix. The **transpose** of A, denoted by A^T, is the $n \times m$ matrix obtained by interchanging the rows and columns of A.

In other words, the entry in the i-th row and j-th column of A becomes the entry in the j-th row and i-th column of A^T. Thus, $(A^T)_{ji} = a_{ij}$.

Example 1.2.2. If

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix},$$

then

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

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Notice that the rows of A become the columns of A^T. Also, the order changes from 2×3 to 3×2.

Example 1.2.3. Let $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$. Then $A^T = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$. In particular, the entries on the main diagonal remain unchanged under transposition.

1.2.1 Algebraic Properties of Transpose

Theorem 1.2.4 (Properties of Transposition). *Let A and B be matrices of appropriate orders over $\mathbb{K}$, and let $\alpha \in \mathbb{K}$. Then:*

(i) $(A^T)^T = A$.

(ii) $(A + B)^T = A^T + B^T$.

(iii) $(\alpha A)^T = \alpha A^T$.

(iv) $(AB)^T = B^T A^T$.

Proof. (i) Transposition interchanges rows and columns. Interchanging them a second time restores the original matrix. Hence, $(A^T)^T = A$.

(ii) Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same order. The (i, j)-th entry of $(A + B)^T$ is $a_{ji} + b_{ji}$, which is also the (i, j)-th entry of $A^T + B^T$. Therefore, $(A + B)^T = A^T + B^T$.

(iii) The (i, j)-th entry of $(\alpha A)^T$ is αa_{ji}, which is also the corresponding entry of αA^T. Hence, $(\alpha A)^T = \alpha A^T$.

(iv) Let $A = (a_{ij})$ be of order $m \times p$ and $B = (b_{jk})$ be of order $p \times n$. The (k, i)-th entry of $(AB)^T$ is

$$\left((AB)^T\right)_{ki} = (AB)_{ik} = \sum_{j=1}^{p} a_{ij} b_{jk}.$$

On the other hand, the (k, i)-th entry of $B^T A^T$ is

$$(B^T A^T)_{ki} = \sum_{j=1}^{p} b_{jk} a_{ij} = \sum_{j=1}^{p} a_{ij} b_{jk}.$$

Since the corresponding entries are equal, $(AB)^T = B^T A^T$.

■

The formula $(AB)^T = B^T A^T$ is called the **reversal law for transposition**. It shows that taking the transpose of a product reverses the order of its factors.

Example 1.2.5. Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}.$$

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We have

$$AB = \begin{bmatrix} 4 & 1 \\ 3 & 0 \end{bmatrix}, \qquad (AB)^T = \begin{bmatrix} 4 & 3 \\ 1 & 0 \end{bmatrix}.$$

Also,

$$B^T A^T = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 1 & 0 \end{bmatrix}.$$

Therefore, $(AB)^T = B^T A^T$, as expected.

Theorem 1.2.6 (General Reversal Law for Transposition). *Let $A_1, A_2, \ldots, A_k$ be matrices whose product is defined. Then*

$$(A_1 A_2 \cdots A_k)^T = A_k^T \cdots A_2^T A_1^T.$$

Proof. The result follows by applying the reversal law $(AB)^T = B^T A^T$ repeatedly. For instance,

$$(ABC)^T = C^T (AB)^T = C^T B^T A^T.$$

The same procedure applies to any finite number of matrices, giving the required result. ■

1.3 Inverse of a Matrix and Its Properties

For a non-zero real number a, its multiplicative inverse is $1/a$, since $a(1/a) = 1$. The inverse of a matrix is defined in a similar way, with the identity matrix taking the place of the number 1.

Definition 1.3.1 (Inverse of a Matrix). A square matrix A of order n is said to be **invertible** or **non-singular** if there exists a square matrix B of the same order such that

$$AB = BA = I_n.$$

The matrix B is called the **inverse** of A and is denoted by A^{-1}. Thus, $AA^{-1} = A^{-1}A = I_n$.

A square matrix that does not possess an inverse is called **singular**.

Example 1.3.2. Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Therefore, $AB = BA = I_2$, and hence

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

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Theorem 1.3.3 (Uniqueness of Inverse). *If a square matrix A is invertible, then its inverse is unique.*

Proof. Suppose B and C are both inverses of A. Then $AB = BA = I_n$ and $AC = CA = I_n$. Using these equalities, we obtain

$$B = BI_n = B(AC) = (BA)C = I_nC = C.$$

Hence, $B = C$, proving that the inverse is unique. ■

Theorem 1.3.4 (Cancellation Law for Invertible Matrices). *Let A be an invertible square matrix, and let B, C be matrices of suitable orders. Then:*

(i) $AB = AC$ *implies* $B = C$.

(ii) $BA = CA$ *implies* $B = C$.

Proof. (i) If $AB = AC$, multiplying both sides on the left by A^{-1} gives $A^{-1}AB = A^{-1}AC$. Thus, $I_nB = I_nC$, and hence $B = C$.

(ii) If $BA = CA$, multiplying both sides on the right by A^{-1} gives $BAA^{-1} = CAA^{-1}$. Therefore, $B = C$.

■

1.3.1 Properties of Inverse and Reversal Laws

Theorem 1.3.5 (Properties of Matrix Inverses). *Let A and B be invertible square matrices of order n over $\mathbb{K}$, and let $\alpha \in \mathbb{K}$ be non-zero. Then:*

(i) $(A^{-1})^{-1} = A$.

(ii) $(\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}$.

(iii) $(AB)^{-1} = B^{-1}A^{-1}$.

(iv) $(A^T)^{-1} = (A^{-1})^T$.

Proof. (i) Since $AA^{-1} = A^{-1}A = I_n$, the matrix A is the inverse of A^{-1}. Hence, $(A^{-1})^{-1} = A$.

(ii) Since $\alpha \neq 0$, we have

$$(\alpha A)\left(\frac{1}{\alpha}A^{-1}\right) = \left(\frac{1}{\alpha}A^{-1}\right)(\alpha A) = I_n.$$

Therefore, $(\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}$.

(iii) We have

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I_n.$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I_n.$$

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Thus, $B^{-1}A^{-1}$ is the inverse of AB, and hence $(AB)^{-1} = B^{-1}A^{-1}$.

(iv) Since $AA^{-1} = A^{-1}A = I_n$, taking transposes and using the reversal law gives

$$(A^{-1})^T A^T = I_n, \qquad A^T (A^{-1})^T = I_n.$$

Therefore, $(A^{-1})^T$ is the inverse of A^T, so $(A^T)^{-1} = (A^{-1})^T$.

■

Example 1.3.6. Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Their inverses are

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \qquad B^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}.$$

Since $AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, we obtain

$$(AB)^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

On the other hand,

$$B^{-1}A^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Thus, $(AB)^{-1} = B^{-1}A^{-1}$.

Theorem 1.3.7 (General Reversal Law for Inverses). *Let $A_1, A_2, \ldots, A_k$ be invertible square matrices of the same order. Then their product is invertible and*

$$(A_1 A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1} A_1^{-1}.$$

Proof. The result follows by repeated application of $(AB)^{-1} = B^{-1}A^{-1}$. For three invertible matrices,

$$(ABC)^{-1} = C^{-1}(AB)^{-1} = C^{-1}B^{-1}A^{-1}.$$

Repeating this procedure gives the stated formula for any finite number of invertible matrices.

■

Example 1.3.8. Let A, B, C be invertible square matrices of the same order. Simplify $(AB^{-1}C)^{-1}$.

Solution. Applying the reversal law, we obtain

$$(AB^{-1}C)^{-1} = C^{-1}(B^{-1})^{-1}A^{-1} = C^{-1}BA^{-1}.$$

■

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1.4 Conjugate Transpose and Its Algebraic Properties

The transpose of a matrix is obtained by interchanging its rows and columns. For matrices with complex entries, another important operation is the **conjugate transpose**, which combines transposition with complex conjugation. Before defining this operation, we recall the meaning of the complex conjugate of a number.

If $z = a + ib$, where $a, b \in \mathbb{R}$ and $i^2 = -1$, then its complex conjugate is $\bar{z} = a - ib$. Thus, complex conjugation changes the sign of the imaginary part while leaving the real part unchanged. For example, $\overline{3+2i} = 3-2i$, $\overline{4-5i} = 4+5i$, and $\overline{7} = 7$.

The following elementary properties will be useful:

$$\overline{z+w} = \bar{z} + \bar{w}, \qquad \overline{zw} = \bar{z}\,\bar{w}, \qquad \bar{\bar{z}} = z.$$

In particular, z is real if and only if $\bar{z} = z$, and z is purely imaginary (including zero) if and only if $\bar{z} = -z$.

Definition 1.4.1 (Conjugate Matrix)**.** Let $A = (a_{ij}) \in M_{m\times n}(\mathbb{C})$. The **conjugate matrix** of A, denoted by $\bar{A}$, is obtained by replacing every entry of A by its complex conjugate. Thus, $\bar{A} = (\bar{a}_{ij})$.

Definition 1.4.2 (Conjugate Transpose)**.** Let $A \in M_{m\times n}(\mathbb{C})$. The **conjugate transpose** of A, denoted by A^*, is the transpose of its conjugate matrix. Thus,

$$A^* = (\bar{A})^T = \overline{A^T}.$$

Equivalently, if $A = (a_{ij})$, then the (j,i)-th entry of A^* is $\bar{a}_{ij}$. The conjugate transpose is also called the **Hermitian transpose**.

To find the conjugate transpose of a matrix, take the complex conjugate of every entry and then interchange the rows and columns. These two operations may also be performed in the reverse order.

Example 1.4.3. Find the conjugate transpose of

$$A = \begin{bmatrix} 1+i & 2-3i & 4 \\ 3i & -2 & 1-i \end{bmatrix}.$$

Solution. First, take the complex conjugate of each entry:

$$\bar{A} = \begin{bmatrix} 1-i & 2+3i & 4 \\ -3i & -2 & 1+i \end{bmatrix}.$$

Now transpose this matrix:

$$\boxed{A^* = \begin{bmatrix} 1-i & -3i \\ 2+3i & -2 \\ 4 & 1+i \end{bmatrix}.}$$

Since A has order 2×3, its conjugate transpose has order 3×2. ■

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1.4.1 Algebraic Properties of Conjugate Transpose

Theorem 1.4.4 (Properties of Conjugate Transposition). *Let A and B be complex matrices of suitable orders, and let $\alpha \in \mathbb{C}$. Then:*

(i) $(A^*)^* = A$.

(ii) $(A+B)^* = A^* + B^*$.

(iii) $(\alpha A)^* = \bar{\alpha} A^*$.

(iv) $(AB)^* = B^* A^*$.

(v) If A is invertible, then $(A^*)^{-1} = (A^{-1})^*$.

Proof. (i) Taking the conjugate transpose twice restores the original matrix, since transposition and complex conjugation both restore the original entries when applied twice. More precisely,

$$(A^*)^* = \overline{(\overline{A^T})^T} = \overline{\bar{A}} = A.$$

(ii) Using the properties of transpose and complex conjugation, we obtain

$$(A+B)^* = \overline{(A+B)^T} = \overline{A^T + B^T} = \overline{A^T} + \overline{B^T} = A^* + B^*.$$

(iii) Since the conjugate of a product of complex numbers is the product of their conjugates,

$$(\alpha A)^* = \overline{(\alpha A)^T} = \overline{\alpha A^T} = \bar{\alpha}\,\overline{A^T} = \bar{\alpha} A^*.$$

(iv) By the reversal law for transposition and the properties of complex conjugation,

$$(AB)^* = \overline{(AB)^T} = \overline{B^T A^T} = \overline{B^T}\,\overline{A^T} = B^* A^*.$$

(v) Suppose A is invertible. Then $AA^{-1} = A^{-1}A = I_n$. Taking conjugate transposes gives

$$(A^{-1})^* A^* = I_n, \qquad A^* (A^{-1})^* = I_n.$$

Therefore, $(A^{-1})^*$ is the inverse of A^*, and hence $(A^*)^{-1} = (A^{-1})^*$.

■

Example 1.4.5. Let

$$A = \begin{bmatrix} 1+i & 2 \\ -i & 3 \end{bmatrix}.$$

Verify that $(iA)^* = -iA^*$.

Solution. First,

$$iA = \begin{bmatrix} -1+i & 2i \\ 1 & 3i \end{bmatrix},$$

so

$$(iA)^* = \begin{bmatrix} -1-i & 1 \\ -2i & -3i \end{bmatrix}.$$

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On the other hand,

$$A^* = \begin{bmatrix} 1-i & i \\ 2 & 3 \end{bmatrix}.$$

Therefore,

$$-iA^* = \begin{bmatrix} -1-i & 1 \\ -2i & -3i \end{bmatrix} = (iA)^*.$$

Hence, the required identity is verified. ■

Example 1.4.6. Let

$$A = \begin{bmatrix} 1 & i \\ 0 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} i & 1 \\ 2 & 0 \end{bmatrix}.$$

Verify that $(AB)^* = B^*A^*$.

Solution. Multiplying the matrices gives

$$AB = \begin{bmatrix} 3i & 1 \\ 4 & 0 \end{bmatrix}.$$

Therefore,

$$(AB)^* = \begin{bmatrix} -3i & 4 \\ 1 & 0 \end{bmatrix}.$$

Also,

$$B^* = \begin{bmatrix} -i & 2 \\ 1 & 0 \end{bmatrix}, \qquad A^* = \begin{bmatrix} 1 & 0 \\ -i & 2 \end{bmatrix}.$$

Thus,

$$B^*A^* = \begin{bmatrix} -i & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -i & 2 \end{bmatrix} = \begin{bmatrix} -3i & 4 \\ 1 & 0 \end{bmatrix}.$$

Hence, $(AB)^* = B^*A^*$. ■

1.5 Properties of Symmetric, Skew-Symmetric, Hermitian, and Skew-Hermitian Matrices

Certain square matrices possess special properties under transposition or conjugate transposition. Four important classes are symmetric, skew-symmetric, Hermitian, and skew-Hermitian matrices. These matrices occur frequently in matrix theory and have useful algebraic properties.

We begin with symmetric and skew-symmetric matrices, which are defined using the ordinary transpose.

1.5.1 Symmetric and Skew-Symmetric Matrices

Definition 1.5.1 (Symmetric Matrix)**.** A square matrix A is called **symmetric** if $A^T = A$. Equivalently, if $A = (a_{ij})$, then $a_{ij} = a_{ji}$ for every i and j.

In a symmetric matrix, the entries on opposite sides of the main diagonal are equal. Thus, the portion above the main diagonal determines the portion below it.

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Example 1.5.2. The matrix

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 3 & 4 & 5 \\ -1 & 5 & 6 \end{bmatrix}$$

is symmetric because

$$A^T = \begin{bmatrix} 2 & 3 & -1 \\ 3 & 4 & 5 \\ -1 & 5 & 6 \end{bmatrix} = A.$$

Definition 1.5.3 (Skew-Symmetric Matrix)**.** A square matrix A is called **skew-symmetric** if $A^T = -A$. Equivalently, if $A = (a_{ij})$, then $a_{ij} = -a_{ji}$ for every i and j.

In a skew-symmetric matrix, the entries on opposite sides of the main diagonal have opposite signs.

Example 1.5.4. Consider

$$A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 4 \\ 3 & -4 & 0 \end{bmatrix}.$$

Then

$$A^T = \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & -4 \\ -3 & 4 & 0 \end{bmatrix} = -A.$$

Hence, A is skew-symmetric.

Theorem 1.5.5. *Every diagonal entry of a skew-symmetric matrix is zero.*

Proof. Let $A = (a_{ij})$ be skew-symmetric. Since $A^T = -A$, we have $a_{ij} = -a_{ji}$ for all i, j. Taking $i = j$, we get $a_{ii} = -a_{ii}$, so $2a_{ii} = 0$. Therefore, $a_{ii} = 0$ for every i. ■

Example 1.5.6. Find x, y, z if

$$A = \begin{bmatrix} 0 & x+1 & 3 \\ 2-y & 0 & z-1 \\ -3 & 4 & 0 \end{bmatrix}$$

is skew-symmetric.

Solution. Since $A^T = -A$, corresponding entries on opposite sides of the main diagonal must have opposite signs. Therefore,

$$x + 1 = -(2 - y), \qquad z - 1 = -4.$$

Thus, $x = y - 3$ and $z = -3$. Hence, the matrix is skew-symmetric precisely when $x = y - 3$ and $z = -3$. ■

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1.5.2 Algebraic Properties of Symmetric and Skew-Symmetric Matrices

Theorem 1.5.7. *If A and B are symmetric matrices of the same order, then $A+B$ and $A-B$ are symmetric.*

Proof. Since $A^T = A$ and $B^T = B$, we have

$$(A+B)^T = A^T + B^T = A + B$$

and $(A-B)^T = A^T - B^T = A - B$. Hence, both $A+B$ and $A-B$ are symmetric. ■

Theorem 1.5.8. *If A and B are skew-symmetric matrices of the same order, then $A+B$ and $A-B$ are skew-symmetric.*

Proof. Since $A^T = -A$ and $B^T = -B$,

$$(A+B)^T = A^T + B^T = -A - B = -(A+B).$$

Similarly, $(A-B)^T = -(A-B)$. Thus, both matrices are skew-symmetric. ■

Theorem 1.5.9 (Product of Symmetric Matrices)**.** *Let A and B be symmetric matrices of the same order. Then AB is symmetric if and only if $AB = BA$.*

Proof. Since $A^T = A$ and $B^T = B$, the reversal law gives

$$(AB)^T = B^T A^T = BA.$$

Therefore, $(AB)^T = AB$ if and only if $BA = AB$. Hence, AB is symmetric precisely when A and B commute. ■

Example 1.5.10. Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Both matrices are symmetric. However,

$$AB = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}, \qquad BA = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}.$$

Since $AB \neq BA$, their product is not symmetric.

Theorem 1.5.11 (Symmetry of AA^T and A^TA)**.** *Let A be any matrix of order $m \times n$ over $\mathbb{K}$. Then AA^T and A^TA are symmetric matrices.*

Proof. Using the reversal law for transposition,

$$(AA^T)^T = (A^T)^T A^T = AA^T.$$

Similarly,

$$(A^TA)^T = A^T(A^T)^T = A^TA.$$

Therefore, both products are symmetric. ■

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Theorem 1.5.12 (Determinant of a Skew-Symmetric Matrix of Odd Order). *Every skew-symmetric matrix of odd order has determinant zero.*

Proof. Let A be a skew-symmetric matrix of odd order n. Since $A^T = -A$, taking determinants gives

$$|A| = |A^T| = |-A| = (-1)^n |A|.$$

Since n is odd, $(-1)^n = -1$. Therefore, $|A| = -|A|$, which implies $2|A| = 0$. Hence, $|A| = 0$. ■

1.5.3 Decomposition into Symmetric and Skew-Symmetric Matrices

An important property of square matrices is that every square matrix can be expressed as the sum of a symmetric matrix and a skew-symmetric matrix. This decomposition provides a convenient way of studying a general matrix through two matrices having special structural properties.

Theorem 1.5.13 (Decomposition of a Square Matrix). *Every square matrix A over $\mathbb{K}$ can be expressed uniquely as*

$$A = S + K,$$

where S is symmetric and K is skew-symmetric. These matrices are given by

$$S = \frac{1}{2}(A + A^T), \qquad K = \frac{1}{2}(A - A^T).$$

Proof. **Existence.** Define $S = \frac{1}{2}(A + A^T)$ and $K = \frac{1}{2}(A - A^T)$. Taking transposes,

$$S^T = \frac{1}{2}(A^T + A) = S,$$

so S is symmetric. Similarly,

$$K^T = \frac{1}{2}(A^T - A) = -K,$$

so K is skew-symmetric. Also,

$$S + K = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = A.$$

Hence, such a decomposition exists.

Uniqueness. Suppose $A = S_1 + K_1$, where S_1 is symmetric and K_1 is skew-symmetric. Taking transposes gives $A^T = S_1 - K_1$. Adding and subtracting these equations, we obtain

$$S_1 = \frac{1}{2}(A + A^T) = S, \qquad K_1 = \frac{1}{2}(A - A^T) = K.$$

Thus, the symmetric and skew-symmetric parts are uniquely determined. ■

Example 1.5.14. Express

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}$$

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as the sum of a symmetric matrix and a skew-symmetric matrix.

Solution. We have $A^T = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$. Therefore, the symmetric part is

$$S = \frac{1}{2}(A + A^T) = \frac{1}{2}\begin{bmatrix} 4 & 6 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}.$$

The skew-symmetric part is

$$K = \frac{1}{2}(A - A^T) = \frac{1}{2}\begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}.$$

Hence,

$$\boxed{A = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}.}$$

The first matrix is symmetric, while the second is skew-symmetric. ■

1.5.4 Hermitian and Skew-Hermitian Matrices

We have already studied symmetric and skew-symmetric matrices, which are defined using the transpose A^T. For matrices with complex entries, two related classes of matrices are defined using the conjugate transpose A^*. These are called **Hermitian** and **skew-Hermitian** matrices.

Definition 1.5.15 (Hermitian Matrix)**.** A square matrix $A \in M_n(\mathbb{C})$ is called **Hermitian** if $A^* = A$. Equivalently, if $A = (a_{ij})$, then $a_{ij} = \bar{a}_{ji}$ for all i, j.

Thus, a matrix is Hermitian if each entry is the complex conjugate of the entry in the corresponding position on the opposite side of the main diagonal.

Example 1.5.16. Consider the matrix

$$A = \begin{bmatrix} 2 & 3+i \\ 3-i & 4 \end{bmatrix}.$$

Its conjugate transpose is

$$A^* = \begin{bmatrix} 2 & 3+i \\ 3-i & 4 \end{bmatrix} = A.$$

Hence, A is Hermitian. Notice that the off-diagonal entries $3+i$ and $3-i$ are complex conjugates of each other, while the diagonal entries are real.

Definition 1.5.17 (Skew-Hermitian Matrix)**.** A square matrix $A \in M_n(\mathbb{C})$ is called **skew-Hermitian** if $A^* = -A$. Equivalently, if $A = (a_{ij})$, then $a_{ij} = -\bar{a}_{ji}$ for all i, j.

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Example 1.5.18. Consider

$$A = \begin{bmatrix} 2i & 3+i \\ -3+i & -4i \end{bmatrix}.$$

Taking the conjugate transpose gives

$$A^* = \begin{bmatrix} -2i & -3-i \\ 3-i & 4i \end{bmatrix} = -A.$$

Therefore, A is skew-Hermitian.

1.5.5 Diagonal Entries and General Forms

One of the simplest ways to recognise Hermitian and skew-Hermitian matrices is to examine their diagonal entries and the relationship between entries on opposite sides of the main diagonal.

Theorem 1.5.19 (Diagonal Entries). *Let $A = (a_{ij})$ be a complex square matrix.*

(i) If A is Hermitian, then every diagonal entry of A is real.

(ii) If A is skew-Hermitian, then every diagonal entry of A is purely imaginary or zero.

Proof. (i) If A is Hermitian, then $a_{ij} = \bar{a}_{ji}$. Taking $i = j$, we obtain $a_{ii} = \bar{a}_{ii}$. Since a complex number equals its conjugate precisely when it is real, every diagonal entry is real.

(ii) If A is skew-Hermitian, then $a_{ij} = -\bar{a}_{ji}$. Taking $i = j$, we obtain $a_{ii} = -\bar{a}_{ii}$. Writing $a_{ii} = x + iy$, where $x, y \in \mathbb{R}$, gives $x + iy = -x + iy$, so $x = 0$. Thus, $a_{ii} = iy$ is purely imaginary or zero.

■

Example 1.5.20. The general form of a Hermitian matrix of order 2 is

$$A = \begin{bmatrix} a & z \\ \bar{z} & b \end{bmatrix}, \qquad a, b \in \mathbb{R}, \quad z \in \mathbb{C}.$$

Similarly, the general form of a skew-Hermitian matrix of order 2 is

$$B = \begin{bmatrix} ia & z \\ -\bar{z} & ib \end{bmatrix}, \qquad a, b \in \mathbb{R}, \quad z \in \mathbb{C}.$$

These forms follow directly from the definitions and provide a convenient method for recognising such matrices.

Example 1.5.21. Find the real numbers x, y, z for which

$$A = \begin{bmatrix} 2 & x+3i \\ 4-yi & z \end{bmatrix}$$

is Hermitian.

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Solution. For A to be Hermitian, we must have $A^* = A$. In particular, its off-diagonal entries must be complex conjugates:

$$x + 3i = \overline{4 - yi} = 4 + yi.$$

Equating real and imaginary parts, we obtain $x = 4$ and $y = 3$. The diagonal entries must be real, which is already satisfied because $2, z \in \mathbb{R}$. Therefore, A is Hermitian when $x = 4$, $y = 3$, and z is any real number. ■

Example 1.5.22. Find the real numbers x, y for which

$$A = \begin{bmatrix} 2i & 3 + xi \\ y + i & -4i \end{bmatrix}$$

is skew-Hermitian.

Solution. Since $A^* = -A$, the off-diagonal entries must satisfy

$$y + i = -\overline{3 + xi} = -3 + xi.$$

Comparing real and imaginary parts gives $y = -3$ and $x = 1$. The diagonal entries $2i$ and $-4i$ are already purely imaginary. Hence, the required values are $x = 1$ and $y = -3$. ■

1.5.6 Algebraic Properties of Hermitian and Skew-Hermitian Matrices

Theorem 1.5.23. *Let A and B be Hermitian matrices of the same order. Then $A + B$ and $A - B$ are Hermitian.*

Proof. Since $A^* = A$ and $B^* = B$, we have

$$(A + B)^* = A^* + B^* = A + B.$$

Similarly, $(A - B)^* = A - B$. Thus, both matrices are Hermitian. ■

Theorem 1.5.24. *Let A and B be skew-Hermitian matrices of the same order. Then $A + B$ and $A - B$ are skew-Hermitian.*

Proof. Since $A^* = -A$ and $B^* = -B$,

$$(A + B)^* = A^* + B^* = -A - B = -(A + B).$$

Similarly, $(A - B)^* = -(A - B)$. Hence, both matrices are skew-Hermitian. ■

Theorem 1.5.25 (Multiplication by i)**.** *Let A be a complex square matrix. Then:*

(i) A is Hermitian if and only if iA is skew-Hermitian.

(ii) A is skew-Hermitian if and only if iA is Hermitian.

Proof. Using $(\alpha A)^* = \bar{\alpha} A^*$ and $\bar{i} = -i$, we have

$$(iA)^* = -iA^*.$$

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If $A^* = A$, then $(iA)^* = -iA = -(iA)$, so iA is skew-Hermitian. Conversely, if $(iA)^* = -(iA)$, then $-iA^* = -iA$, which gives $A^* = A$. This proves (i).

Similarly, if $A^* = -A$, then $(iA)^* = -i(-A) = iA$, so iA is Hermitian. Conversely, if $(iA)^* = iA$, then $-iA^* = iA$, giving $A^* = -A$. This proves (ii). ■

Example 1.5.26. Let

$$A = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix}.$$

Since $A^* = A$, it is Hermitian. Multiplying by i gives

$$iA = \begin{bmatrix} 2i & -1+i \\ 1+i & 3i \end{bmatrix}.$$

Taking the conjugate transpose, we obtain $(iA)^* = -iA$. Hence, iA is skew-Hermitian.

Theorem 1.5.27 (Product of Hermitian Matrices)**.** *Let A and B be Hermitian matrices of the same order. Then AB is Hermitian if and only if $AB = BA$.*

Proof. Since $A^* = A$ and $B^* = B$, the reversal law gives

$$(AB)^* = B^*A^* = BA.$$

Therefore, $(AB)^* = AB$ if and only if $BA = AB$. Hence, AB is Hermitian precisely when A and B commute. ■

Theorem 1.5.28 (Hermitian Products AA^* and A^*A)**.** *Let $A \in M_{m\times n}(\mathbb{C})$. Then both AA^* and A^*A are Hermitian.*

Proof. Using the reversal law and $(A^*)^* = A$, we obtain

$$(AA^*)^* = (A^*)^*A^* = AA^*$$

and

$$(A^*A)^* = A^*(A^*)^* = A^*A.$$

Hence, both products are Hermitian. ■

Example 1.5.29. Let

$$A = \begin{bmatrix} 1 & i \\ 2 & 0 \end{bmatrix}.$$

Then

$$A^* = \begin{bmatrix} 1 & 2 \\ -i & 0 \end{bmatrix}.$$

Consequently,

$$AA^* = \begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix}, \qquad A^*A = \begin{bmatrix} 5 & i \\ -i & 1 \end{bmatrix}.$$

Both matrices are Hermitian, as their conjugate transposes equal themselves.

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1.5.7 Decomposition into Hermitian and Skew-Hermitian Parts

Just as every square matrix can be written as the sum of a symmetric matrix and a skew-symmetric matrix, every complex square matrix can be expressed as the sum of a Hermitian matrix and a skew-Hermitian matrix.

Theorem 1.5.30 (Hermitian and Skew-Hermitian Decomposition)**.** *Every complex square matrix A can be expressed uniquely as*

$$A = H + K,$$

where H is Hermitian and K is skew-Hermitian. These matrices are given by

$$H = \frac{1}{2}(A + A^*), \qquad K = \frac{1}{2}(A - A^*).$$

Proof. **Existence.** Define $H = \frac{1}{2}(A + A^*)$ and $K = \frac{1}{2}(A - A^*)$. Taking conjugate transposes,

$$H^* = \frac{1}{2}(A^* + A) = H,$$

so H is Hermitian. Similarly,

$$K^* = \frac{1}{2}(A^* - A) = -K,$$

so K is skew-Hermitian. Moreover,

$$H + K = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*) = A.$$

Hence, the required decomposition exists.

Uniqueness. Suppose $A = H_1 + K_1$, where H_1 is Hermitian and K_1 is skew-Hermitian. Taking the conjugate transpose gives $A^* = H_1 - K_1$. Adding and subtracting these equations yields

$$H_1 = \frac{1}{2}(A + A^*) = H, \qquad K_1 = \frac{1}{2}(A - A^*) = K.$$

Therefore, the decomposition is unique. ■

1.5.8 Cartesian Decomposition of a Complex Matrix

Every complex number $z = a + ib$ can be expressed uniquely in terms of its real and imaginary parts, where $a, b \in \mathbb{R}$. A similar decomposition is possible for every complex square matrix, with Hermitian matrices taking the place of real numbers.

Theorem 1.5.31 (Cartesian Decomposition)**.** *Every complex square matrix A can be expressed uniquely in the form*

$$A = P + iQ,$$

where both P and Q are Hermitian matrices. They are given by

$$P = \frac{1}{2}(A + A^*), \qquad Q = \frac{1}{2i}(A - A^*).$$

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Proof. **Existence.** Define $P = \frac{1}{2}(A + A^*)$ and $Q = \frac{1}{2i}(A - A^*)$. We first show that both matrices are Hermitian. Since $(A^*)^* = A$,

$$P^* = \frac{1}{2}(A^* + A) = P.$$

Also, using $\overline{\frac{1}{2i}} = -\frac{1}{2i}$, we obtain

$$Q^* = -\frac{1}{2i}(A^* - A) = \frac{1}{2i}(A - A^*) = Q.$$

Thus, both P and Q are Hermitian. Moreover,

$$P + iQ = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*) = A.$$

Hence, the decomposition exists.

Uniqueness. Suppose $A = P_1 + iQ_1$, where P_1 and Q_1 are Hermitian. Taking the conjugate transpose gives $A^* = P_1 - iQ_1$. Adding and subtracting these equations, we obtain

$$P_1 = \frac{1}{2}(A + A^*) = P, \qquad Q_1 = \frac{1}{2i}(A - A^*) = Q.$$

Hence, the decomposition is unique. ■

The representation $A = P + iQ$ requires P and Q to be Hermitian. Since iQ is skew-Hermitian, it is equivalent to the form $A = H + K$ with H Hermitian and K skew-Hermitian. The two forms should not be confused.

Example 1.5.32. Express

$$A = \begin{bmatrix} 2+3i & 1-i \\ 3+5i & 4 \end{bmatrix}$$

in the form $A = P + iQ$, where P and Q are Hermitian.

Solution. First, calculate

$$A^* = \begin{bmatrix} 2-3i & 3-5i \\ 1+i & 4 \end{bmatrix}.$$

The first Hermitian component is

$$P = \frac{1}{2}(A + A^*) = \frac{1}{2}\begin{bmatrix} 4 & 4-6i \\ 4+6i & 8 \end{bmatrix} = \begin{bmatrix} 2 & 2-3i \\ 2+3i & 4 \end{bmatrix}.$$

The second Hermitian component is

$$Q = \frac{1}{2i}(A - A^*) = \frac{1}{2i}\begin{bmatrix} 6i & -2+4i \\ 2+4i & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2+i \\ 2-i & 0 \end{bmatrix}.$$

Both P and Q are Hermitian. Hence,

$$\boxed{A = \begin{bmatrix} 2 & 2-3i \\ 2+3i & 4 \end{bmatrix} + i\begin{bmatrix} 3 & 2+i \\ 2-i & 0 \end{bmatrix}.}$$

■

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1.5.9 Comparison of Special Classes of Matrices

The defining conditions and important structural properties of the four classes of matrices studied above may be summarised as follows:

Type of Matrix	Defining Condition	Diagonal Entries
Symmetric	$A^T = A$	Unrestricted
Skew-symmetric	$A^T = -A$	Zero
Hermitian	$A^* = A$	Real
Skew-Hermitian	$A^* = -A$	Purely imaginary or zero

The essential distinction is that symmetric and skew-symmetric matrices are defined using ordinary transposition, while Hermitian and skew-Hermitian matrices are defined using conjugate transposition.

1.6 Adjoint of a Square Matrix and Matrix Inversion

The classical adjoint $\text{adj}(A)$, defined using cofactors, is different from the conjugate transpose A^*.

The inverse of a square matrix plays an important role in matrix algebra, particularly in solving systems of linear equations. One method of finding the inverse involves the **adjoint** of the matrix. To understand this method, we first recall the concepts of minors and cofactors, which are defined using determinants.

1.6.1 Minors and Cofactors

Definition 1.6.1 (Minor of an Element)**.** Let $A = (a_{ij})$ be a square matrix of order $n \geq 2$. The **minor** of the element a_{ij}, denoted by M_{ij}, is the determinant of the matrix obtained by deleting the i-th row and j-th column of A.

Example 1.6.2. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

To find the minor M_{12}, delete the first row and second column. The remaining entries form the matrix $\begin{bmatrix} 4 & 6 \\ 7 & 9 \end{bmatrix}$. Therefore,

$$M_{12} = \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = 4(9) - 6(7) = -6.$$

Similarly, deleting the second row and third column gives

$$M_{23} = \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = 8 - 14 = -6.$$

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Definition 1.6.3 (Cofactor of an Element)**.** Let M_{ij} be the minor of the element a_{ij} of a square matrix A. The **cofactor** of a_{ij}, denoted by C_{ij}, is defined by

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Thus, the cofactor is the minor taken with a positive or negative sign according as $i + j$ is even or odd.

For a matrix of order 3, the signs of the cofactors follow the pattern

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

The signs alternate along every row and column, beginning with a positive sign in the first position.

Example 1.6.4. For the matrix A in the preceding example, we found $M_{12} = -6$ and $M_{23} = -6$. Hence,

$$C_{12} = (-1)^{1+2} M_{12} = -(-6) = 6,$$

and

$$C_{23} = (-1)^{2+3} M_{23} = -(-6) = 6.$$

Notice that the minor and cofactor of an element need not be equal.

1.6.2 Adjoint of a Square Matrix

Definition 1.6.5 (Adjoint of a Matrix)**.** Let $A = (a_{ij})$ be a square matrix of order $n \geq 2$, and let C_{ij} denote the cofactor of a_{ij}. The matrix

$$C = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$$

is called the **cofactor matrix** of A. The **adjoint** of A, denoted by $\text{adj}(A)$, is the transpose of its cofactor matrix. Thus,

$$\text{adj}(A) = C^T = (C_{ji})_{n \times n}.$$

The adjoint of a square matrix is therefore calculated in three steps:

(i) Find the minor of each element of the matrix.

(ii) Apply the appropriate signs to obtain the cofactors.

(iii) Form the cofactor matrix and take its transpose.

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Example 1.6.6. Find the adjoint of $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$.

Solution. The minors of the four elements are $M_{11} = 4$, $M_{12} = 1$, $M_{21} = 3$, and $M_{22} = 2$. Applying the cofactor signs, we obtain

$$C_{11} = 4, \quad C_{12} = -1, \quad C_{21} = -3, \quad C_{22} = 2.$$

Thus, the cofactor matrix is

$$C = \begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix}.$$

Taking its transpose, we obtain

$$\operatorname{adj}(A) = C^T = \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}.$$

■

For a matrix of order 2, the adjoint has a particularly simple form.

Theorem 1.6.7 (Adjoint of a 2×2 Matrix). *If* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, *then*

$$\operatorname{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Proof. The cofactors of the elements of A are $C_{11} = d$, $C_{12} = -c$, $C_{21} = -b$, and $C_{22} = a$. Therefore, the cofactor matrix is $\begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$. Taking its transpose gives

$$\operatorname{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

■

Example 1.6.8. Find the adjoint of the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution. We first calculate the cofactors of all nine elements. Expanding the corresponding determinants of order 2, we get

$$C_{11} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1, \qquad C_{12} = -\begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix} = 0, \qquad C_{13} = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0,$$

$$C_{21} = -\begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = -2, \quad C_{22} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1, \qquad C_{23} = -\begin{vmatrix} 1 & 2 \\ 0 & 0 \end{vmatrix} = 0,$$

$$C_{31} = \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2, \qquad C_{32} = -\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = -1, \quad C_{33} = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1.$$

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Hence, the cofactor matrix is

$$C = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix}.$$

Taking the transpose gives

$$\boxed{\operatorname{adj}(A) = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.}$$

■

1.6.3 Fundamental Property of the Adjoint

The importance of the adjoint arises from its relationship with the determinant of a matrix. This relationship provides a direct method for finding the inverse of a non-singular matrix.

We recall that the determinant of a square matrix can be expanded along any row or column by multiplying each entry by its corresponding cofactor and adding the products. For example, expansion along the first row gives $\det(A) = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$.

Theorem 1.6.9 (Fundamental Property of the Adjoint)**.** *Let A be a square matrix of order n. Then*

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = |A| I_n,$$

where $|A|$ denotes the determinant of A.

Proof. Let $A = (a_{ij})$ and let C_{ij} denote the cofactor of a_{ij}. Since the (j,k)-th entry of $\operatorname{adj}(A)$ is C_{kj}, the (i,k)-th entry of $A \operatorname{adj}(A)$ is

$$\big(A \operatorname{adj}(A)\big)_{ik} = \sum_{j=1}^{n} a_{ij} C_{kj}.$$

We consider two cases.

Case 1: $i = k$**.** In this case,

$$\sum_{j=1}^{n} a_{ij} C_{ij} = |A|,$$

by expansion of the determinant along the i-th row.

Case 2: $i \neq k$**.** Consider the matrix obtained from A by replacing its k-th row with its i-th row. This matrix has two equal rows, so its determinant is zero. Expanding that determinant along the k-th row gives

$$\sum_{j=1}^{n} a_{ij} C_{kj} = 0.$$

Thus, every diagonal entry of $A \operatorname{adj}(A)$ equals $|A|$, while every off-diagonal entry is zero. Therefore,

$$A \operatorname{adj}(A) = |A| I_n.$$

Similarly, by expanding determinants along columns, we obtain $\operatorname{adj}(A)A = |A| I_n$. Combining these results gives

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = |A| I_n.$$

■

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1.6.4 Connection Between Adjoint and Matrix Inversion

We have already defined the inverse of a square matrix A as a matrix A^{-1} satisfying $AA^{-1} = A^{-1}A = I_n$. The fundamental property of the adjoint enables us to calculate this inverse whenever the determinant of A is non-zero.

Theorem 1.6.10 (Condition for Invertibility). *A square matrix A is invertible if and only if $|A| \neq 0$. Moreover, when $|A| \neq 0$, its inverse is given by*

$$\boxed{A^{-1} = \frac{1}{|A|}\operatorname{adj}(A).}$$

Proof. Suppose first that $|A| \neq 0$. By the fundamental property of the adjoint,

$$A\operatorname{adj}(A) = \operatorname{adj}(A)A = |A|I_n.$$

Dividing both sides by $|A|$, we obtain

$$A\left(\frac{1}{|A|}\operatorname{adj}(A)\right) = \left(\frac{1}{|A|}\operatorname{adj}(A)\right)A = I_n.$$

Hence, A is invertible and

$$A^{-1} = \frac{1}{|A|}\operatorname{adj}(A).$$

Conversely, suppose that A is invertible. Then $AA^{-1} = I_n$. Taking determinants of both sides gives

$$|A|\,|A^{-1}| = |I_n| = 1.$$

Therefore, $|A| \neq 0$. Hence, a square matrix is invertible if and only if its determinant is non-zero. ■

1.6.5 Finding the Inverse by the Adjoint Method

The preceding theorem gives a systematic procedure for finding the inverse of a square matrix.

1. Calculate the determinant $|A|$. If $|A| = 0$, the matrix has no inverse.
2. If $|A| \neq 0$, calculate all the cofactors of A.
3. Form the cofactor matrix and take its transpose to obtain $\operatorname{adj}(A)$.
4. Apply the formula $A^{-1} = \dfrac{1}{|A|}\operatorname{adj}(A)$.

Example 1.6.11. Find the inverse of $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$ using the adjoint method.

Solution. First, calculate the determinant:

$$|A| = 3(1) - 2(1) = 1 \neq 0.$$

Hence, A is invertible. Interchanging the diagonal entries and changing the signs of the off-diagonal entries gives

$$\operatorname{adj}(A) = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}.$$

Therefore,

$$\boxed{A^{-1} = \frac{1}{|A|}\operatorname{adj}(A) = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}.}$$

■

Example 1.6.12. Find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

using the adjoint method.

Solution. Since A is an upper triangular matrix, its determinant is the product of its diagonal entries. Hence, $|A| = 1(1)(1) = 1 \neq 0$, so A is invertible.

We have already calculated its adjoint:

$$\operatorname{adj}(A) = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore,

$$\boxed{A^{-1} = \frac{1}{|A|}\operatorname{adj}(A) = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.}$$

To verify the result, we calculate

$$AA^{-1} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

Thus, the calculated matrix is indeed the inverse of A. ■

Example 1.6.13. Determine whether the matrix $A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ is invertible.

Solution. We calculate $|A| = 2(2) - 4(1) = 0$. Since the determinant is zero, A is singular. Therefore, A^{-1} does not exist. ■

Exercise 1.6.14. Solve the following exercises:

1. Construct two 2×2 real matrices A and B such that $AB = 0$ (the zero matrix) but $BA \neq 0$. What does this imply about zero divisors in matrix algebra?

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2. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$. Compute $(A+B)^2$ and $A^2 + 2AB + B^2$. Show that $(A+B)^2 \neq A^2 + 2AB + B^2$, and state the necessary and sufficient condition under which equality holds.

3. Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ be matrices over $\mathbb{C}$. Compute AB and BA, and show that A and B anti-commute (i.e., $AB = -BA$).

4. Verify the reversal law for transposition $(AB)^T = B^T A^T$ for the matrices:
$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ -1 & 1 \end{bmatrix}.$$

5. Using the reversal law for two matrices, prove by induction that for any k conformable matrices $A_1, A_2, \ldots, A_k$:
$$(A_1 A_2 \cdots A_k)^T = A_k^T \cdots A_2^T A_1^T.$$

6. For $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$, find A^{-1} and explicitly verify that $AA^{-1} = A^{-1}A = I_2$.

7. Given $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, calculate $(AB)^{-1}$ and $B^{-1}A^{-1}$, and confirm that $(AB)^{-1} = B^{-1}A^{-1}$.

8. Show that if $A \in M_n(\mathbb{K})$ is an invertible matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$. Verify this result for $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$.

9. Let $A = \begin{bmatrix} 2+i & 3-2i \\ 4i & 1+5i \end{bmatrix} \in M_2(\mathbb{C})$. Find the conjugate transpose A^*, and verify that $(2iA)^* = -2iA^*$.

10. Let $A \in M_{m\times p}(\mathbb{C})$ and $B \in M_{p\times n}(\mathbb{C})$. Prove formally from definition that $(AB)^* = B^*A^*$.

11. Express the real matrix $A = \begin{bmatrix} 3 & 8 & 2 \\ 2 & 5 & 6 \\ 4 & 0 & 1 \end{bmatrix}$ uniquely in the form $A = S + K$, where S is symmetric ($S^T = S$) and K is skew-symmetric ($K^T = -K$).

12. Prove that if A is an $n \times n$ skew-symmetric matrix and n is an odd integer, then $\det(A) = 0$. Using this property, state $\det(A)$ without expansion for:
$$A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}.$$

13. Prove that all main diagonal entries of a Hermitian matrix are real numbers, whereas all main diagonal entries of a skew-Hermitian matrix are either zero or purely imaginary.

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14. Show that if H is a Hermitian matrix, then $K = iH$ is a skew-Hermitian matrix. Conversely, show that if K is skew-Hermitian, then $H = -iK$ is Hermitian.

15. Decompose the complex matrix $A = \begin{bmatrix} 1+2i & 3-i \\ 2+4i & 5i \end{bmatrix}$ into the form $A = P + iQ$, where P is a Hermitian matrix and Q is a skew-Hermitian matrix.

16. Find the cofactor matrix C and the adjoint matrix $\text{adj}(A)$ for:
$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}.$$

17. For $A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}$, compute $\text{adj}(A)$ and verify the fundamental identity $A \cdot \text{adj}(A) = \det(A) I_3$.

18. Let A be a 3×3 non-singular matrix with $\det(A) = 4$.
    (a) Prove that $\det(\text{adj}(A)) = (\det(A))^2$.
    (b) Find the numerical value of $\det(\text{adj}(A))$.

19. Prove that for any non-singular matrix $A \in M_n(\mathbb{K})$ ($n \geq 2$):
$$\text{adj}(\text{adj}(A)) = (\det(A))^{n-2} A.$$

20. Using the adjoint method ($A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$), determine the inverse of the matrix:
$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}.$$

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CHAPTER

2

ELEMENTARY OPERATIONS AND SYSTEMS OF LINEAR EQUATIONS

Elementary operations simplify matrices by changing one row or column at a time. In a system of equations, elementary row operations replace equations by equivalent equations and therefore preserve the solution set. Row reduction allows us to identify pivots, decide whether a square matrix is invertible, and find its inverse.

We begin with elementary operations and their matrix representations, then study echelon and normal forms and the factorization of invertible matrices. Finally, we apply row reduction to homogeneous and non-homogeneous systems, including systems whose solutions depend on a parameter.

2.1 Elementary Operations and Elementary Matrices

2.1.1 Elementary Row Operations

Definition 2.1.1 (Elementary Row Operations)**.** An **elementary row operation** is an operation on the rows of a matrix of one of the following three types:

(i) **Interchange of rows:** $R_i \leftrightarrow R_j$, which interchanges the i-th and j-th rows.

(ii) **Multiplication of a row by a non-zero scalar:** $R_i \to cR_i$, where $c \neq 0$.

(iii) **Addition of a multiple of one row to another:** $R_i \to R_i + cR_j$, where $i \neq j$.

Here R_i denotes the i-th row of the matrix.

Elementary row operations are particularly useful for creating zero entries below or above a chosen non-zero entry. This allows a matrix to be reduced systematically to a simpler form.

Example 2.1.2. Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. The three elementary row operations can be illustrated as follows:

(i) Interchanging the rows gives

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}.$$

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(ii) Multiplying the second row by 2 gives

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{R_2 \to 2R_2} \begin{bmatrix} 1 & 2 \\ 6 & 8 \end{bmatrix}.$$

(iii) Subtracting three times the first row from the second gives

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 3R_1} \begin{bmatrix} 1 & 2 \\ 3-3(1) & 4-3(2) \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}.$$

Each elementary row operation can be reversed by another elementary row operation. For example, the inverse of $R_i \to cR_i$ is $R_i \to c^{-1}R_i$, and the inverse of $R_i \to R_i + cR_j$ is $R_i \to R_i - cR_j$. A row interchange is reversed by performing the same interchange again.

Definition 2.1.3 (Row Equivalent Matrices)**.** Two matrices A and B of the same order are said to be **row equivalent** if one can be obtained from the other by a finite sequence of elementary row operations. We write $A \sim B$.

2.1.2 Elementary Matrices

Definition 2.1.4 (Elementary Matrix)**.** An **elementary matrix** is a square matrix obtained by performing exactly one elementary row operation on an identity matrix.

Example 2.1.5. Starting with the identity matrix I_3, we obtain the following elementary matrices:

$$R_1 \leftrightarrow R_2: \quad E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$R_2 \to 3R_2: \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$R_3 \to R_3 - 2R_1: \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

Each matrix is obtained by performing the indicated operation on I_3.

An important property of elementary matrices is that multiplying a matrix on the left by an elementary matrix performs the corresponding elementary row operation.

Theorem 2.1.6 (Representation of Elementary Row Operations)**.** *Let A be an $m \times n$ matrix, and let E be the elementary matrix obtained by performing an elementary row operation on I_m. Then EA is the matrix obtained by performing the same row operation on A.*

Proof. Let the rows of A be denoted by $R_1, R_2, \ldots, R_m$. Since $I_m A = A$, each row of A is obtained by multiplying the corresponding row of I_m by A.

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If two rows of I_m are interchanged, the corresponding rows of the product are interchanged. If the i-th row of I_m is multiplied by $c \neq 0$, the i-th row of the product becomes cR_i. Similarly, replacing the i-th row of I_m by itself plus c times its j-th row replaces R_i by $R_i + cR_j$ in the product.

Thus, in each case, multiplying A on the left by E performs the corresponding elementary row operation. ■

Example 2.1.7. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Represent the row operation $R_2 \to R_2 - 3R_1$ by an elementary matrix and verify that both methods give the same result.

Solution. Applying the given row operation to I_2 gives

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 3R_1} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} = E.$$

Multiplying A on the left by E, we obtain

$$EA = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}.$$

Now perform the row operation directly on A:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 3R_1} \begin{bmatrix} 1 & 2 \\ 3 - 3(1) & 4 - 3(2) \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}.$$

The two methods give the same matrix. Hence, multiplication by E on the left produces exactly the same result as the corresponding elementary row operation. ■

Theorem 2.1.8 (Invertibility of Elementary Matrices)**.** *Every elementary matrix is invertible, and its inverse is also an elementary matrix.*

Proof. Let E be an elementary matrix representing an elementary row operation. Since every elementary row operation can be reversed by another elementary row operation, let F represent the inverse operation.

Applying the two operations successively to I_n restores I_n. Hence, $FE = EF = I_n$, so $E^{-1} = F$. Since F is elementary, the inverse of an elementary matrix is also elementary. ■

Example 2.1.9. Find the inverse of $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$.

Solution. The matrix E represents $R_3 \to R_3 - 2R_1$. Its inverse operation is $R_3 \to R_3 + 2R_1$. Therefore,

$$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}.$$

■

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2.1.3 Elementary Column Operations

Elementary column operations are defined in the same manner as elementary row operations. They consist of interchanging two columns, multiplying a column by a non-zero scalar, and adding a multiple of one column to another.

If C_i denotes the i-th column of a matrix, these operations are represented by $C_i \leftrightarrow C_j$, $C_i \to cC_i$ with $c \neq 0$, and $C_i \to C_i + cC_j$ with $i \neq j$.

Theorem 2.1.10 (Representation of Elementary Column Operations)**.** *Let A be an $m \times n$ matrix, and let F be obtained by performing an elementary column operation on I_n. Then AF is the matrix obtained by performing the same column operation on A.*

Proof. Since $AI_n = A$, the columns of A are unchanged by multiplication on the right by I_n. An elementary column operation changes the columns of I_n by interchange, scaling, or replacement. Multiplication AF then makes exactly the corresponding change to the columns of A. ■

Example 2.1.11. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Represent the column operation $C_2 \to C_2 - 2C_1$ by an elementary matrix.

Solution. Applying the operation to I_2 gives

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{C_2 \to C_2 - 2C_1} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = F.$$

Thus,

$$AF = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix}.$$

Applying the operation directly also gives

$$A \xrightarrow{C_2 \to C_2 - 2C_1} \begin{bmatrix} 1 & 2-2(1) \\ 3 & 4-2(3) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix}.$$

Therefore, both methods give the same result. ■

Elementary row operations are represented by multiplication on the left, whereas column operations are represented by multiplication on the right. Both preserve rank. When solving $AX = B$, row operations on the augmented matrix preserve its solutions. Column operations on A generally change the unknowns and must be accompanied by a change of variables.

2.2 Row Echelon Form and Row Reduction

Elementary row operations are used to reduce a matrix to a form in which the leading entries of successive non-zero rows move progressively to the right. Such a form makes it easier to identify the number of independent equations and to solve systems of linear equations.

Definition 2.2.1 (Row Echelon Form)**.** A matrix is said to be in **row echelon form** (REF) if:

(i) All zero rows, if any, occur below the non-zero rows.

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(ii) The first non-zero entry of each non-zero row lies strictly to the right of the first non-zero entry of the preceding row.

The first non-zero entry of a non-zero row is called its **leading entry** or **pivot**.

The second condition also ensures that all entries below each pivot are zero.

Example 2.2.2. The matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

are in row echelon form. However, the matrix

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \\ 0 & 5 & 6 \end{bmatrix}$$

is not in row echelon form because the leading entry of the third row is to the left of the leading entry of the second row.

Definition 2.2.3 (Reduced Row Echelon Form)**.** A matrix is in **reduced row echelon form** (RREF) if:

(i) It is in row echelon form.

(ii) Every pivot is equal to 1.

(iii) Every pivot is the only non-zero entry in its column.

For example,

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

is in reduced row echelon form, whereas

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

is in row echelon form but not in reduced row echelon form.

The procedure of obtaining row echelon form by elementary row operations is called **Gaussian elimination**. Continuing the reduction until reduced row echelon form is obtained is called **Gauss–Jordan elimination**.

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2.2.1 Reduction to Row Echelon Form

Example 2.2.4. Reduce $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ 3 & 8 & 7 \end{bmatrix}$ to row echelon form.

Solution. We first make the entries below the leading entry $a_{11} = 1$ equal to zero. Applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$, we obtain

$$A \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{bmatrix}.$$

The leading entry of the second row is now 1. To make the entry below it zero, apply $R_3 \to R_3 - 2R_2$:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

The final matrix is in row echelon form because the zero row is at the bottom and the pivot of the second row lies to the right of the pivot of the first row. ■

2.2.2 Reduction to Reduced Row Echelon Form

Example 2.2.5. Reduce $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \\ 1 & 2 & 3 \end{bmatrix}$ to reduced row echelon form.

Solution. Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$A \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Next, apply $R_3 \to R_3 - R_2$ to make the entry below the second pivot zero:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

This matrix is in row echelon form. To obtain reduced row echelon form, we must also make the entry above the second pivot zero. Applying $R_1 \to R_1 - R_2$, we obtain

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The final matrix is in reduced row echelon form, since each pivot is 1 and is the only non-zero entry in its column. ■

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Definition 2.2.6 (Rank of a Matrix)**.** The **rank** of a matrix A, denoted by $\text{rank}(A)$, is the number of pivots in any row echelon form of A.

Thus, the rank of a matrix can be determined by reducing it to row echelon form and counting its non-zero rows. Elementary row operations do not change the rank of a matrix.

In both preceding examples, the reduced matrices have two non-zero rows. Hence, the rank of each original matrix is 2. The interpretation of rank in terms of linearly independent rows and columns will be studied in the next unit.

2.3 Normal Form of a Matrix

Row echelon form is obtained using elementary row operations. When both elementary row and column operations are allowed, a matrix can be reduced to an even simpler form called its **normal form**.

Definition 2.3.1 (Normal Form of a Matrix)**.** Let A be an $m \times n$ matrix of rank r. The matrix

$$N = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}_{m \times n}$$

is called the **normal form** or **rank canonical form** of A.

Unlike RREF, which may have non-zero entries in non-pivot columns, normal form contains an identity block and zeros elsewhere. Its identity block has size equal to the rank. For example, the normal form of a 3×4 matrix of rank 2 is

$$N = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Theorem 2.3.2 (Reduction to Normal Form)**.** *Let A be an $m \times n$ matrix of rank r. Then there exist non-singular matrices $P \in M_m(\mathbb{K})$ and $Q \in M_n(\mathbb{K})$ such that*

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}_{m \times n}.$$

Proof. By elementary row operations, A can be reduced to its reduced row echelon form R. Suppose the rank is r. Then R has exactly r pivot columns, each containing a single 1 and zeros elsewhere.

By interchanging columns, move the pivot columns to the first r positions, keeping their original relative order. The resulting matrix has the form

$$\begin{bmatrix} I_r & B \\ 0 & 0 \end{bmatrix}.$$

Now subtract suitable multiples of the first r columns from each remaining column to make the entries of B zero. This gives

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

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Every elementary row operation is represented by multiplication on the left by an invertible elementary matrix. Likewise, every elementary column operation is represented by multiplication on the right by an invertible elementary matrix.

Let P and Q be the respective products of these elementary matrices. Since products of invertible matrices are invertible, P and Q are non-singular. Therefore,

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

■

Example 2.3.3. Find non-singular matrices P and Q such that PAQ is in normal form for

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 1 & 3 & 8 \end{bmatrix}.$$

Solution. We first use row operations to obtain an echelon form. Apply $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$, and then $R_3 \to R_3 - 2R_2$:

$$A \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 2 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} = R.$$

Performing these operations in the same order on I_3 gives

$$P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}, \qquad PA = R.$$

Now eliminate the entries outside the identity block by column operations: $C_2 \to C_2 - C_1$, $C_3 \to C_3 - 2C_1$, and $C_3 \to C_3 - 3C_2$:

$$R \longrightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Performing these column operations on I_3 gives

$$Q = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus P and Q are invertible, and

$$\boxed{PAQ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.}$$

The normal form contains I_2, so $\operatorname{rank}(A) = 2$. ■

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2.4 Invertibility and Row Reduction

Elementary row operations provide an alternative to the adjoint method for determining whether a square matrix is invertible. They also give a systematic method for calculating its inverse.

Theorem 2.4.1 (Invertibility and Row Reduction). *Let A be a square matrix of order n. Then the following statements are equivalent:*

(i) *A is invertible.*

(ii) *A is row equivalent to I_n.*

(iii) *The reduced row echelon form of A is I_n.*

(iv) $\operatorname{rank}(A) = n$.

Proof. Suppose first that A is row equivalent to I_n. Then there exist elementary matrices $E_1, E_2, \ldots, E_k$ such that $E_k \cdots E_2 E_1 A = I_n$. Since every elementary matrix is invertible,

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1},$$

so A is invertible.

Conversely, suppose A is invertible. Then $|A| \neq 0$. Elementary row operations preserve the property of having a non-zero determinant. Hence, the row echelon form of A cannot contain a zero row. It must therefore have n pivots, and further row operations reduce it to I_n.

Thus, A is invertible if and only if it is row equivalent to I_n. Since an $n \times n$ matrix has reduced row echelon form I_n precisely when it has n pivots, the remaining equivalences follow. ■

Example 2.4.2. Determine whether $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ is invertible using elementary row operations.

Solution. Applying $R_2 \to R_2 - 3R_1$, we obtain

$$A \sim \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}.$$

Next, apply $R_2 \to -R_2$ and $R_1 \to R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Since A is row equivalent to I_2, it is invertible. ■

2.4.1 Finding the Inverse by Row Reduction

Theorem 2.4.3 (Inverse by Elementary Row Operations). *Let A be an invertible square matrix of order n. Its inverse can be found by applying elementary row operations to the augmented*

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matrix $[A \mid I_n]$ until the left-hand block becomes I_n. The resulting matrix is

$$[A \mid I_n] \sim [I_n \mid A^{-1}].$$

Proof. Since A is invertible, elementary row operations reduce it to I_n. Let the corresponding elementary matrices be $E_1, E_2, \ldots, E_k$. Then

$$E_k E_{k-1} \cdots E_1 A = I_n.$$

Multiplying on the right by A^{-1} gives $E_k E_{k-1} \cdots E_1 = A^{-1}$. Therefore, applying the same row operations to I_n produces A^{-1}, proving the result. ■

Example 2.4.4. Find the inverse of $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ by elementary row operations.

Solution. Start with the augmented matrix

$$[A \mid I_2] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right].$$

Applying $R_2 \to R_2 - 3R_1$, we obtain

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right].$$

Next, apply $R_2 \to -\frac{1}{2}R_2$:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right].$$

Finally, apply $R_1 \to R_1 - 2R_2$:

$$\left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right].$$

Since the left-hand block has become I_2, the right-hand block is A^{-1}. Therefore,

$$\boxed{A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}}.$$

■

2.5 Non-Singular Matrices as Products of Elementary Matrices

Elementary matrices not only represent row operations but also provide a way of expressing any invertible square matrix as a product of simpler matrices.

Theorem 2.5.1 (Elementary Matrix Factorization). *A square matrix is non-singular if and only if it can be expressed as a finite product of elementary matrices.*

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Proof. Suppose A is a non-singular square matrix of order n. Since A is invertible, it can be reduced to I_n by elementary row operations. Hence, there exist elementary matrices $E_1, E_2, \ldots, E_k$ such that

$$E_k E_{k-1} \cdots E_1 A = I_n.$$

Multiplying successively by their inverses, we obtain

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1}.$$

Since the inverse of every elementary matrix is elementary, A is a product of elementary matrices.

Conversely, suppose $A = E_1 E_2 \cdots E_k$, where each E_i is elementary. Every elementary matrix is invertible, and a product of invertible matrices is invertible. Hence, A is non-singular. ■

Example 2.5.2. Express $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ as a product of elementary matrices.

Solution. First, reduce A to I_2. Apply $R_2 \to \frac{1}{3}R_2$:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Next, apply $R_1 \to R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The corresponding elementary matrices are

$$E_1 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix}, \qquad E_2 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Since $E_2 E_1 A = I_2$, we obtain $A = E_1^{-1} E_2^{-1}$. Therefore,

$$\boxed{A = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.}$$

Both factors are elementary matrices, and their product equals A. ■

2.6 Systems of Linear Equations and Their Solution Sets

A system of linear equations consists of one or more linear equations involving a common set of unknowns. It may have one solution, infinitely many solutions, or no solution. By applying elementary row operations to the equations together, we obtain an equivalent system that is easier to solve.

Consider a system of m linear equations in n unknowns:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

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This system can be written in the compact matrix form $AX = B$, where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

Here A is called the **coefficient matrix**, X is the column matrix of unknowns, and B is the column matrix of constants. The matrix obtained by adjoining B to A is called the **augmented matrix** and is denoted by $[A \mid B]$.

Example 2.6.1. Write the system $x + 2y - z = 3$, $2x - y + 3z = 5$ in matrix form.

Solution. The coefficient matrix, column matrix of unknowns, and column matrix of constants are

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

Hence, the system is $AX = B$, and its augmented matrix is

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & 5 \end{array}\right].$$

■

2.6.1 Homogeneous Systems of Linear Equations

Definition 2.6.2 (Homogeneous System)**.** A system of linear equations is called **homogeneous** if all its constant terms are zero. In matrix form, a homogeneous system is written as $AX = 0$.

Every homogeneous system has at least one solution, namely $X = 0$, obtained by assigning zero to every unknown. This is called the **trivial solution**. Any solution in which at least one unknown is non-zero is called a **non-trivial solution**.

Example 2.6.3. The system $x + 2y - z = 0$, $2x + 4y - 2z = 0$ is homogeneous. Clearly, $x = y = z = 0$ is a solution. However, $x = -2$, $y = 1$, and $z = 0$ also satisfy both equations, so the system has a non-trivial solution.

To solve a homogeneous system, we reduce its coefficient matrix to row echelon form or reduced row echelon form. The variables corresponding to pivot columns are called **pivot variables**, while the remaining variables are called **free variables**. The free variables may be assigned arbitrary values, and the pivot variables are then determined from the equations.

Theorem 2.6.4 (Non-Trivial Solutions of Homogeneous Systems)**.** *Let A be an $m \times n$ matrix of rank r. Then the homogeneous system $AX = 0$ has a non-trivial solution if and only if $r < n$. In particular, $AX = 0$ has only the trivial solution if and only if $r = n$.*

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Proof. Reduce A to reduced row echelon form. Since $\text{rank}(A) = r$, there are r pivot variables and $n - r$ free variables.

If $r = n$, every variable is a pivot variable, so the homogeneous system has only the solution $X = 0$.

If $r < n$, at least one free variable is present. Assigning a non-zero value to one free variable and zero to the others determines a solution in which at least one variable is non-zero. Hence, the system has a non-trivial solution. ■

Corollary 2.6.5. *A homogeneous system having more unknowns than equations always has a non-trivial solution.*

Proof. If the system has m equations and n unknowns with $n > m$, then $\text{rank}(A) \leq m < n$. Hence, by the preceding theorem, the system has a non-trivial solution. ■

2.6.2 Solution of Homogeneous Systems by Row Reduction

Example 2.6.6. Solve the homogeneous system

$$x + 2y - z = 0, \qquad 2x + 5y + z = 0, \qquad 3x + 7y = 0.$$

Solution. The coefficient matrix is

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 5 & 1 \\ 3 & 7 & 0 \end{bmatrix}.$$

Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{bmatrix}.$$

Next, apply $R_3 \to R_3 - R_2$ and $R_1 \to R_1 - 2R_2$:

$$A \sim \begin{bmatrix} 1 & 0 & -7 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

The corresponding equations are $x - 7z = 0$ and $y + 3z = 0$. The variable z is free. Taking $z = t$, where $t \in \mathbb{K}$, we obtain $x = 7t$ and $y = -3t$. Hence, the general solution is

$$\boxed{X = t \begin{bmatrix} 7 \\ -3 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{K}.}$$

The system has infinitely many solutions because t can take any value in $\mathbb{K}$. The trivial solution is obtained by taking $t = 0$, while every $t \neq 0$ gives a non-trivial solution. ■

Example 2.6.7. Solve the homogeneous system $x + y + z + w = 0$, $2x + y + 3z + 2w = 0$.

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Solution. The coefficient matrix is

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 3 & 2 \end{bmatrix}.$$

Applying $R_2 \to R_2 - 2R_1$, $R_2 \to -R_2$, and $R_1 \to R_1 - R_2$, we obtain

$$A \sim \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 0 \end{bmatrix}.$$

Thus, $x + 2z + w = 0$ and $y - z = 0$, giving $x = -2z - w$ and $y = z$.

Since z and w are free variables, let $z = s$ and $w = t$. Then $x = -2s - t$ and $y = s$. Therefore,

$$\boxed{X = s \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{K}.}$$

There are two arbitrary parameters because the coefficient matrix has rank 2 and the system has four unknowns. ■

2.7 Non-Homogeneous Systems and Consistency

Definition 2.7.1 (Non-Homogeneous System)**.** A system $AX = B$ is called **non-homogeneous** if $B \neq 0$, that is, if at least one constant term is non-zero.

Unlike a homogeneous system, a non-homogeneous system need not possess a solution.

Definition 2.7.2 (Consistent and Inconsistent Systems)**.** A system of linear equations is called **consistent** if it has at least one solution. It is called **inconsistent** if it has no solution.

A consistent system may have exactly one solution or infinitely many solutions. An inconsistent system has no solution.

To determine consistency, we reduce the augmented matrix $[A \mid B]$ by elementary row operations. Each row of the augmented matrix represents an equation, and elementary row operations replace the system by an equivalent system having the same solution set.

A row operation on an augmented matrix changes the equations but not the solution set. In particular, a row of the form

$$\left[\begin{array}{cccc|c} 0 & 0 & \cdots & 0 & c \end{array}\right], \qquad c \neq 0,$$

represents the impossible equation $0 = c$. Its occurrence shows that the system is inconsistent.

Example 2.7.3. Consider the system $x + y = 2$, $2x + 2y = 5$. Its augmented matrix reduces as

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 5 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 1 \end{array}\right].$$

The second row represents $0 = 1$, which is impossible. Hence, the system is inconsistent.

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2.7.1 Condition for Consistency

Theorem 2.7.4 (Rouché–Capelli Theorem). *Let $AX = B$ be a system of m linear equations in n unknowns. The system is consistent if and only if*

$$\boxed{\operatorname{rank}(A) = \operatorname{rank}([A \mid B]).}$$

Furthermore, if the common rank is r, then:

(i) the system has a unique solution when $r = n$;

(ii) the system has infinitely many solutions when $r < n$.

Proof. Apply elementary row operations to the augmented matrix $[A \mid B]$ until the coefficient part is in row echelon form. Since the same operations are applied to both parts, the resulting equations have exactly the same solutions as the original system.

Suppose $\operatorname{rank}(A) < \operatorname{rank}([A \mid B])$. The additional pivot of the augmented matrix must occur in its last column. In echelon form this yields a row of the form $[\,0 \;\cdots\; 0 \mid c\,]$ with $c \neq 0$, corresponding to the impossible equation $0 = c$. Hence, the system is inconsistent.

Conversely, suppose $\operatorname{rank}(A) = \operatorname{rank}([A \mid B]) = r$. No contradictory equation occurs. We may assign arbitrary values to the free variables and determine the pivot variables successively. Thus, the system is consistent.

If $r = n$, every variable is a pivot variable, so there are no free variables and the solution is unique. If $r < n$, there are $n - r$ free variables, which may take arbitrary values. Since $\mathbb{K}$ is either $\mathbb{R}$ or $\mathbb{C}$, the system then has infinitely many solutions. ■

The consistency criterion may be summarised as follows:

$$\begin{aligned}
&\operatorname{rank}(A) < \operatorname{rank}([A \mid B]) && \Longrightarrow \text{ no solution,}\\
&\operatorname{rank}(A) = \operatorname{rank}([A \mid B]) = n && \Longrightarrow \text{ unique solution,}\\
&\operatorname{rank}(A) = \operatorname{rank}([A \mid B]) < n && \Longrightarrow \text{ infinitely many solutions.}
\end{aligned}$$

Example 2.7.5. Solve the system

$$x + y + z = 6, \qquad 2x - y + z = 3, \qquad x + 2y - z = 2.$$

Solution. The augmented matrix is

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array}\right].$$

Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -4 \end{array}\right].$$

Interchange R_2 and R_3, and then apply $R_3 \to R_3 + 3R_2$:

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array}\right].$$

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The final matrix represents

$$x + y + z = 6, \qquad y - 2z = -4, \qquad -7z = -21.$$

From the last equation, $z = 3$. Substituting into the second equation gives $y = 2$, and substituting these values into the first gives $x = 1$. Therefore,

$$\boxed{X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.}$$

Since $\text{rank}(A) = \text{rank}([A \mid B]) = 3$, the system has a unique solution. ■

Example 2.7.6. Find the general solution of the system

$$x + y + z = 2, \qquad 2x + 2y + 2z = 4, \qquad x - y + z = 0.$$

Solution. The augmented matrix is

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 4 \\ 1 & -1 & 1 & 0 \end{array}\right].$$

Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & -2 \end{array}\right].$$

Interchanging the last two rows, multiplying the second row by $-\frac{1}{2}$, and applying $R_1 \to R_1 - R_2$, we obtain

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Thus, $x + z = 1$ and $y = 1$. Let $z = t$, where $t \in \mathbb{K}$. Then $x = 1 - t$ and $y = 1$. Hence,

$$\boxed{X = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{K}.}$$

Since $\text{rank}(A) = \text{rank}([A \mid B]) = 2 < 3$, the system has infinitely many solutions. ■

2.7.2 Systems Involving a Parameter

Some systems contain an unknown constant or parameter, such as λ or k. In such cases, the number of solutions may depend on the value of the parameter. Row reduction helps us determine the values for which the system is consistent and whether its solution is unique.

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Example 2.7.7. Determine the values of λ for which the system

$$x+y+z=1, \qquad 2x+2y+2z=\lambda, \qquad x-y+z=0$$

is consistent. Find its general solution when it is consistent.

Solution. The augmented matrix is

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & \lambda \\ 1 & -1 & 1 & 0 \end{array}\right].$$

Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & \lambda - 2 \\ 0 & -2 & 0 & -1 \end{array}\right].$$

The second row represents $0 = \lambda - 2$. Therefore, the system is consistent if and only if $\lambda = 2$.

For $\lambda = 2$, the remaining equations are $x+y+z=1$ and $-2y=-1$. Hence, $y=\frac{1}{2}$ and $x+z=\frac{1}{2}$.

Taking $z=t$, we obtain $x=\frac{1}{2}-t$. Therefore,

$$\boxed{X = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{K}.}$$

Thus, the system has infinitely many solutions when $\lambda = 2$ and no solution when $\lambda \neq 2$. ■

For a square system, the determinant provides a convenient way of identifying parameter values for which a unique solution exists.

Theorem 2.7.8 (Unique Solution of a Square System). *Let A be a square matrix of order n. The system $AX = B$ has a unique solution for every $B \in \mathbb{K}^n$ if and only if A is non-singular. In this case, the unique solution is $X = A^{-1}B$.*

Proof. Suppose A is non-singular. Multiplying $AX = B$ by A^{-1} on the left gives $X = A^{-1}B$, which is therefore a solution. If X_1 and X_2 are two solutions, then $A(X_1 - X_2) = 0$. Multiplying by A^{-1} gives $X_1 = X_2$, so the solution is unique.

Conversely, suppose $AX = B$ has a unique solution for every $B \in \mathbb{K}^n$. In particular, $AX = 0$ has only the trivial solution. Therefore, $\operatorname{rank}(A) = n$, which implies that A is non-singular. ■

Example 2.7.9. Determine the values of λ for which the system

$$x+y+z=1, \qquad x+\lambda y+2z=2, \qquad x+2y+\lambda z=3$$

has a unique solution. Also determine what happens for the remaining values of λ.

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Solution. The coefficient matrix is

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & 2 \\ 1 & 2 & \lambda \end{bmatrix}.$$

A square system has a unique solution if and only if its coefficient matrix is non-singular. Subtracting the first row from the second and third rows, we obtain

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 0 & \lambda - 1 & 1 \\ 0 & 1 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 - 1 = \lambda(\lambda - 2).$$

Hence, the system has a unique solution whenever $\lambda \neq 0, 2$.

For $\lambda = 0$, row reduction of the augmented matrix gives

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 3 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 1 \\ 0 & 1 & -1 & 2 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{array}\right].$$

The last row represents $0 = 3$, so the system is inconsistent.

For $\lambda = 2$, we obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 2 & 3 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

This system is also inconsistent. Therefore,

$\lambda \neq 0, 2$	: unique solution,
$\lambda = 0$ or 2	: no solution.

■

2.8 Relationship Between Homogeneous and Non-Homogeneous Solutions

Every consistent non-homogeneous system is closely related to its associated homogeneous system. This relationship enables us to describe the complete solution set of a non-homogeneous system once one particular solution is known.

Theorem 2.8.1 (General Solution of a Consistent System). *Suppose the system $AX = B$ is consistent, and let X_p be one particular solution. Then every solution of $AX = B$ can be written as*

$$\boxed{X = X_p + X_h,}$$

where X_h is a solution of the associated homogeneous system $AX = 0$. Conversely, every vector of this form is a solution of $AX = B$.

Proof. Let X be any solution of $AX = B$, and let X_p be a particular solution. Since $AX = B$ and $AX_p = B$, subtraction gives

$$A(X - X_p) = AX - AX_p = B - B = 0.$$

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Thus, $X - X_p$ is a solution of the homogeneous system $AX = 0$. Writing $X_h = X - X_p$, we obtain $X = X_p + X_h$.

Conversely, suppose $AX_p = B$ and $AX_h = 0$. Then

$$A(X_p + X_h) = AX_p + AX_h = B + 0 = B.$$

Hence, $X_p + X_h$ is a solution of $AX = B$. ■

Corollary 2.8.2. *If $AX = B$ is consistent, A has n columns, and $\operatorname{rank}(A) = r$, then its general solution contains $n - r$ arbitrary parameters.*

Proof. The associated homogeneous system $AX = 0$ has $n - r$ free variables. By the preceding theorem, every solution of $AX = B$ is obtained by adding a fixed particular solution to a solution of $AX = 0$. Hence, the general solution contains $n - r$ arbitrary parameters. ■

Example 2.8.3. Express the general solution of

$$x + y + z = 3, \qquad 2x + 2y + 2z = 6$$

as the sum of a particular solution and the general solution of the associated homogeneous system.

Solution. The second equation is twice the first, so the system reduces to $x + y + z = 3$. Taking $y = z = 0$ gives the particular solution

$$X_p = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}.$$

The associated homogeneous system reduces to $x + y + z = 0$. Let $y = s$ and $z = t$. Then $x = -s - t$, so its general solution is

$$X_h = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore, the general solution of the original system is

$$\boxed{X = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{K}.}$$

This expresses every solution as a particular solution plus a solution of the associated homogeneous system. ■

Exercise 2.8.4. Solve the following exercises.

1. Write down the 3×3 elementary matrices corresponding to the following elementary row operations:

   (a) Interchanging the first and third rows ($R_1 \leftrightarrow R_3$).

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(b) Multiplying the second row by -4 ($R_2 \to -4R_2$).

(c) Adding 3 times the second row to the first row ($R_1 \to R_1 + 3R_2$).

2. Prove that every elementary matrix is non-singular and that its inverse is also an elementary matrix of the same type. Write down the inverses of the elementary matrices obtained in Problem 1.

3. Reduce the following matrix A to Row Echelon Form (REF) using elementary row operations and determine its rank:

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -1 & 7 \\ -1 & -2 & 3 & -1 \end{bmatrix}.$$

4. Reduce the following matrix A to its Reduced Row Echelon Form (RREF):

$$A = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 0 & 4 & 8 \end{bmatrix}.$$

5. Find non-singular matrices P and Q such that PAQ is in the normal canonical form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ for:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 1 & 3 & 8 \end{bmatrix}.$$

State the rank r of matrix A.

6. Find the normal form PAQ for the matrix:

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}.$$

7. Using elementary row operations on the augmented matrix $[A \mid I_3]$, compute A^{-1} for:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}.$$

8. Prove that a square matrix $A \in M_n(\mathbb{K})$ is non-singular if and only if A is row-equivalent to the identity matrix I_n.

9. Express the non-singular matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$ as a product of elementary matrices.

10. Express $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ as a product of elementary matrices.

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11. Find the complete solution set of the homogeneous system $AX = 0$. Identify the pivot variables, the free variables, and the arbitrary parameters:

$$\begin{aligned} x_1 + 2x_2 - 3x_3 + x_4 &= 0, \\ 2x_1 + 5x_2 - 8x_3 + 6x_4 &= 0, \\ 3x_1 + 4x_2 - 5x_3 - 4x_4 &= 0. \end{aligned}$$

12. Determine the values of k for which the following homogeneous system has non-trivial solutions, and find the corresponding solutions:

$$\begin{aligned} (1-k)x_1 + 2x_2 + 3x_3 &= 0, \\ 2x_1 + (4-k)x_2 + 6x_3 &= 0, \\ 3x_1 + 6x_2 + (9-k)x_3 &= 0. \end{aligned}$$

13. Show that a homogeneous system $AX = 0$ with m equations and n variables has a non-trivial solution if $n > m$. If $\text{rank}(A) = r$, how many free variables occur in the general solution?

14. Test the consistency of the following non-homogeneous system using rank criteria ($\text{rank}(A) = \text{rank}([A \mid B])$), and solve if consistent:

$$\begin{aligned} x_1 + x_2 + x_3 &= 6, \\ x_1 + 2x_2 + 3x_3 &= 10, \\ x_1 + 2x_2 + 4x_3 &= 11. \end{aligned}$$

15. Investigate for what values of λ and μ the system:

$$\begin{aligned} x + y + z &= 6, \\ x + 2y + 3z &= 10, \\ x + 2y + \lambda z &= \mu, \end{aligned}$$

    has: (i) no solution, (ii) a unique solution, (iii) infinitely many solutions.

16. Solve the following system completely, or show that it is inconsistent:

$$\begin{aligned} x_1 - 2x_2 + x_3 + x_4 &= 2, \\ 3x_1 + 0x_2 + 2x_3 + 2x_4 &= 7, \\ 2x_1 - 4x_2 + 2x_3 + 2x_4 &= 4, \\ 4x_1 - 2x_2 + 3x_3 + 3x_4 &= 8. \end{aligned}$$

17. Suppose X_p is a particular solution of a consistent system $AX = B$. Prove that its complete solution set consists of all vectors $X_p + X_h$, where X_h runs through the solutions of $AX = 0$.

18. Solve the following system of linear equations using Gauss-Jordan elimination:

$$\begin{aligned} 2x_1 + x_2 + 5x_3 &= 4, \\ x_1 + 3x_2 + 2x_3 &= 7, \\ 3x_1 + 4x_2 + 7x_3 &= 11. \end{aligned}$$

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19. Given $A \in M_n(\mathbb{K})$, suppose there exist non-singular matrices P and Q such that $PAQ = I_n$. Show that A is invertible and express A^{-1} explicitly in terms of P and Q.

20. Test whether $AX = B$ is consistent, where $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}$. Explain what your conclusion means about expressing B as a linear combination of the columns of A.

CHAPTER

3

LINEAR COMBINATIONS, RANK AND TRACE

In the preceding chapter, we studied elementary operations and their applications to systems of linear equations. We now use these ideas to investigate linear combinations of vectors, spanning sets, and linear independence. These concepts lead to the rank of a matrix and the dimension of the solution space of a homogeneous system. We conclude with the trace of a square matrix.

Throughout this chapter, $\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$. The set $\mathbb{K}^n$ consists of all column vectors with n entries from $\mathbb{K}$. Thus,

$$\mathbb{K}^n = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} : x_1, \ldots, x_n \in \mathbb{K} \right\}.$$

In particular, $\mathbb{R}^n$ consists of vectors with real entries, while $\mathbb{C}^n$ consists of vectors with complex entries.

3.1 Linear Combinations and Span

3.1.1 Linear Combinations

Definition 3.1.1 (Linear Combination)**.** Let $v_1, v_2, \ldots, v_k \in \mathbb{K}^n$. A vector $v \in \mathbb{K}^n$ is called a **linear combination** of $v_1, v_2, \ldots, v_k$ if there exist scalars $c_1, c_2, \ldots, c_k \in \mathbb{K}$ such that

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k.$$

The scalars $c_1, c_2, \ldots, c_k$ are called the **coefficients** of the linear combination.

A linear combination is obtained by multiplying each given vector by a scalar and then adding the resulting vectors. The scalars may be positive, negative, or zero when working over $\mathbb{R}$, and may also be complex when working over $\mathbb{C}$.

To determine whether a given vector v is a linear combination of $v_1, \ldots, v_k$, we seek scalars $c_1, \ldots, c_k$ satisfying $c_1 v_1 + \cdots + c_k v_k = v$. Equating corresponding entries gives a system of linear equations in the unknown coefficients. If this system is consistent, the required linear combination exists.

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Example 3.1.2. Express $v = \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix}$ as a linear combination of

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Solution. Suppose $v = c_1v_1 + c_2v_2 + c_3v_3$. Then

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix}.$$

Equating corresponding entries gives

$$c_1 + c_3 = 3, \qquad c_1 + c_2 = 5, \qquad c_2 + c_3 = 4.$$

From the first two equations, $c_3 = 3 - c_1$ and $c_2 = 5 - c_1$. Substituting into the third equation gives $(5 - c_1) + (3 - c_1) = 4$, so $c_1 = 2$. Consequently, $c_2 = 3$ and $c_3 = 1$.

Therefore,

$$\boxed{v = 2v_1 + 3v_2 + v_3.}$$

■

3.1.2 Linear Combinations and Matrix Equations

The preceding method can also be expressed using matrix multiplication.

Theorem 3.1.3 (Matrix Criterion for Linear Combinations). *Let $v_1, v_2, \ldots, v_k \in \mathbb{K}^n$ and let*

$$A = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix}.$$

Then a vector $v \in \mathbb{K}^n$ is a linear combination of $v_1, \ldots, v_k$ if and only if the system $AC = v$ is consistent, where $C = \begin{bmatrix} c_1 & c_2 & \cdots & c_k \end{bmatrix}^T$.

Proof. By matrix multiplication,

$$AC = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{bmatrix} = c_1v_1 + c_2v_2 + \cdots + c_kv_k.$$

Thus, $AC = v$ has a solution precisely when v can be expressed as a linear combination of the given vectors. ■

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3.1.3 Linear Span

Definition 3.1.4 (Linear Span)**.** Let $S = \{v_1, v_2, \ldots, v_k\}$ be a set of vectors in $\mathbb{K}^n$. The **linear span** of S, denoted by $\text{span}(S)$ or $\text{span}\{v_1, \ldots, v_k\}$, is the set of all linear combinations of the vectors in S. Thus,

$$\text{span}(S) = \{c_1v_1 + \cdots + c_kv_k : c_1, \ldots, c_k \in \mathbb{K}\}.$$

The span of a collection of vectors consists of every vector that can be obtained by multiplying the given vectors by scalars and adding the results. In particular, the zero vector always belongs to the span, since it is obtained by taking all coefficients equal to zero.

Definition 3.1.5 (Spanning Set)**.** A set of vectors $S \subseteq \mathbb{K}^n$ is said to **span** $\mathbb{K}^n$ if every vector in $\mathbb{K}^n$ can be expressed as a linear combination of vectors from S. Equivalently, $\text{span}(S) = \mathbb{K}^n$.

The distinction between a linear combination and a spanning set is important. A linear combination expresses one particular vector using the given vectors, whereas a spanning set allows us to express every vector in the space.

Example 3.1.6. Show that the vectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

span $\mathbb{R}^3$.

Solution. Let $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be an arbitrary vector in $\mathbb{R}^3$. We seek scalars c_1, c_2, c_3 such that $c_1v_1 + c_2v_2 + c_3v_3 = v$. Equating corresponding entries gives

$$c_1 + c_3 = x, \qquad c_1 + c_2 = y, \qquad c_2 + c_3 = z.$$

Solving these equations, we obtain

$$c_1 = \frac{x + y - z}{2}, \qquad c_2 = \frac{-x + y + z}{2}, \qquad c_3 = \frac{x - y + z}{2}.$$

These scalars exist for every $x, y, z \in \mathbb{R}$. Therefore, every vector in $\mathbb{R}^3$ is a linear combination of v_1, v_2, v_3. Hence,

$$\boxed{\text{span}\{v_1, v_2, v_3\} = \mathbb{R}^3.}$$

■

Theorem 3.1.7 (Matrix Criterion for Spanning)**.** *Let $v_1, \ldots, v_k \in \mathbb{K}^n$, and let $A = \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}$. Then the vectors $v_1, \ldots, v_k$ span $\mathbb{K}^n$ if and only if* $\text{rank}(A) = n$.

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Proof. The vectors span $\mathbb{K}^n$ precisely when every vector $b \in \mathbb{K}^n$ can be expressed as $b = c_1v_1 + \cdots + c_kv_k$. By matrix multiplication, this is equivalent to the system $AC = b$ being consistent for every $b \in \mathbb{K}^n$.

If $\text{rank}(A) = n$, row reduction gives a pivot in every row, so $AC = b$ is consistent for every b. Conversely, if $\text{rank}(A) < n$, row reduction leaves at least one zero row in the coefficient matrix. Since the corresponding row operations are invertible, a suitable choice of b produces a non-zero entry in that row of the augmented matrix, making the system inconsistent. Hence, the vectors span $\mathbb{K}^n$ if and only if $\text{rank}(A) = n$. ■

3.2 Linear Dependence and Independence

A set of vectors may contain unnecessary vectors in the sense that one of them can be expressed as a linear combination of the others. The ideas of linear dependence and independence allow us to identify such situations.

Definition 3.2.1 (Linear Dependence and Independence)**.** Let $v_1, v_2, \ldots, v_k \in \mathbb{K}^n$.

(i) The vectors are called **linearly dependent** if there exist scalars $c_1, \ldots, c_k$, not all zero, such that $c_1v_1 + \cdots + c_kv_k = 0$.

(ii) The vectors are called **linearly independent** if $c_1v_1 + \cdots + c_kv_k = 0$ implies $c_1 = c_2 = \cdots = c_k = 0$.

The equation $c_1v_1 + \cdots + c_kv_k = 0$ always has the solution $c_1 = \cdots = c_k = 0$, called the **trivial solution**. The vectors are linearly dependent precisely when this equation has a non-trivial solution.

The most direct method for testing linear independence is to form the equation $c_1v_1 + \cdots + c_kv_k = 0$ and determine whether the coefficients must all be zero.

Example 3.2.2. Determine whether

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

are linearly independent.

Solution. Suppose $c_1v_1 + c_2v_2 + c_3v_3 = 0$. Then

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Equating corresponding entries gives

$$c_1 + c_3 = 0, \qquad c_1 + c_2 = 0, \qquad c_2 + c_3 = 0.$$

From the first two equations, $c_3 = -c_1$ and $c_2 = -c_1$. Substituting into the third gives $-2c_1 = 0$, so $c_1 = 0$. Hence, $c_2 = c_3 = 0$.

Since only the trivial relation is possible, the vectors are linearly independent. ■

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The preceding method can be carried out systematically by placing the given vectors as columns of a matrix and using elementary row operations.

Theorem 3.2.3 (Matrix Criterion for Linear Independence). *Let $v_1, v_2, \ldots, v_k \in \mathbb{K}^n$ and let*

$$A = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix}.$$

Then the vectors $v_1, \ldots, v_k$ are linearly independent if and only if $\operatorname{rank}(A) = k$.

Proof. Let $C = \begin{bmatrix} c_1 & c_2 & \cdots & c_k \end{bmatrix}^T$. By matrix multiplication,

$$AC = c_1v_1 + c_2v_2 + \cdots + c_kv_k.$$

Hence, the vectors are linearly independent precisely when the homogeneous system $AC = 0$ has only the trivial solution.

From the theory of homogeneous systems, $AC = 0$ has only the trivial solution if and only if every column of A contains a pivot. Since A has k columns, this is equivalent to $\operatorname{rank}(A) = k$. ■

Theorem 3.2.4 (Size Criterion for Linear Dependence). *Any collection of more than n vectors in $\mathbb{K}^n$ is linearly dependent.*

Proof. Let $v_1, \ldots, v_k \in \mathbb{K}^n$, where $k > n$, and form the matrix $A = \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}$. Since A has n rows, $\operatorname{rank}(A) \leq n < k$. Therefore, by the preceding theorem, the vectors are linearly dependent. ■

3.3 Basis and Dimension

The concepts of spanning and linear independence are brought together in the notion of a basis. A basis consists of enough vectors to express every vector in a space, but contains no redundant vectors.

Definition 3.3.1 (Basis). A set $\mathcal{B} = \{v_1, v_2, \ldots, v_k\} \subseteq \mathbb{K}^n$ is called a **basis** of $\mathbb{K}^n$ if:

(i) $\mathcal{B}$ spans $\mathbb{K}^n$;

(ii) $\mathcal{B}$ is linearly independent.

For example, the vectors

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

form a basis of $\mathbb{R}^3$. Indeed, every vector $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ can be expressed as $ae_1 + be_2 + ce_3$, and the equation $c_1e_1 + c_2e_2 + c_3e_3 = 0$ implies $c_1 = c_2 = c_3 = 0$.

More generally, the vectors $e_1, e_2, \ldots, e_n$, where e_i has 1 in its i-th position and zero elsewhere, form the **standard basis** of $\mathbb{K}^n$.

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Theorem 3.3.2 (Basis Criterion). *Let $v_1, v_2, \ldots, v_n \in \mathbb{K}^n$, and let $A = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}$. Then the following statements are equivalent:*

(i) $\{v_1, \ldots, v_n\}$ *is a basis of* $\mathbb{K}^n$.

(ii) The vectors $v_1, \ldots, v_n$ *are linearly independent.*

(iii) The vectors $v_1, \ldots, v_n$ *span* $\mathbb{K}^n$.

(iv) $\operatorname{rank}(A) = n$.

Proof. By the matrix criterion for linear independence, the vectors $v_1, \ldots, v_n$ are linearly independent if and only if $\operatorname{rank}(A) = n$. By the matrix criterion for spanning, they span $\mathbb{K}^n$ if and only if $\operatorname{rank}(A) = n$.

Thus, linear independence and spanning are equivalent for a collection of exactly n vectors in $\mathbb{K}^n$. Since a basis is a linearly independent spanning set, all four statements are equivalent. ■

Example 3.3.3. Determine whether

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

form a basis of $\mathbb{R}^3$.

Solution. Place the vectors as columns of a matrix:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Since

$$|A| = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2 \neq 0,$$

the matrix is invertible and $\operatorname{rank}(A) = 3$. Hence, the vectors are linearly independent and span $\mathbb{R}^3$.

Therefore, $\{v_1, v_2, v_3\}$ is a basis of $\mathbb{R}^3$. ■

Theorem 3.3.4 (Number of Vectors in a Basis). *Every basis of $\mathbb{K}^n$ contains exactly n vectors.*

Proof. Suppose $\mathcal{B} = \{v_1, \ldots, v_k\}$ is a basis of $\mathbb{K}^n$. Since the vectors are linearly independent, the size criterion for linear independence gives $k \leq n$.

On the other hand, since $\mathcal{B}$ spans $\mathbb{K}^n$, the matrix $A = \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}$ has rank n. A matrix cannot have more pivots than columns, so $n \leq k$.

Combining these inequalities gives $k = n$. ■

Definition 3.3.5 (Dimension). The **dimension** of a vector space having a finite basis is the number of vectors in any basis. In particular, since every basis of $\mathbb{K}^n$ contains exactly n

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vectors, we have $\dim(\mathbb{K}^n) = n$.

Thus, $\dim(\mathbb{R}^2) = 2$, $\dim(\mathbb{R}^3) = 3$, and $\dim(\mathbb{C}^n) = n$ when $\mathbb{C}^n$ is considered over the field $\mathbb{C}$.

An important consequence of linear independence is that the representation of a vector in terms of basis vectors is unique.

Theorem 3.3.6 (Unique Representation). *Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis of $\mathbb{K}^n$. Then every vector $x \in \mathbb{K}^n$ can be expressed uniquely in the form*

$$x = c_1v_1 + c_2v_2 + \cdots + c_nv_n.$$

Proof. Since $\mathcal{B}$ spans $\mathbb{K}^n$, every vector $x \in \mathbb{K}^n$ has at least one representation as a linear combination of the basis vectors.

Suppose

$$x = c_1v_1 + \cdots + c_nv_n = d_1v_1 + \cdots + d_nv_n.$$

Subtracting these expressions gives

$$(c_1 - d_1)v_1 + \cdots + (c_n - d_n)v_n = 0.$$

Since the basis vectors are linearly independent, $c_i - d_i = 0$ for every i. Thus, $c_i = d_i$ for all i, proving uniqueness. ■

3.4 Rank of a Matrix

In Chapter 2, rank was introduced as the number of pivots in a row echelon form. We now study its relationship with linear independence and the rows and columns of a matrix.

3.4.1 Row Space and Column Space

Definition 3.4.1 (Row Space and Column Space). Let $A \in M_{m\times n}(\mathbb{K})$.

(i) The **row space** of A is the set of all linear combinations of its rows.

(ii) The **column space** of A is the set of all linear combinations of its columns.

The rows of A are vectors in $\mathbb{K}^n$, while its columns are vectors in $\mathbb{K}^m$. Thus, the row space is a subspace of $\mathbb{K}^n$, and the column space is a subspace of $\mathbb{K}^m$.

The **row rank** of A is the maximum number of linearly independent rows, while the **column rank** is the maximum number of linearly independent columns.

Theorem 3.4.2 (Equality of Row Rank and Column Rank). *For every matrix A, the row rank and column rank are equal.*

Proof. Let A be an $m \times n$ matrix, and suppose its row echelon form is R, having r non-zero rows.

First, we show that the row rank of A is r. Elementary row operations do not change the span of the rows, since each operation can be reversed. Thus, A and R have the same row space.

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The r non-zero rows of R are linearly independent. Indeed, in any linear relation among them, the first pivot column forces the coefficient of the first row to be zero, the second pivot column then forces the coefficient of the second row to be zero, and so on. Hence, the row rank of A is r.

Next, we consider the columns. Since R is obtained from A by elementary row operations, there is an invertible matrix P such that $R = PA$. For any scalars $c_1, \ldots, c_n$, we have

$$c_1 R_1 + \cdots + c_n R_n = P(c_1 A_1 + \cdots + c_n A_n),$$

where A_j and R_j denote the corresponding columns of A and R. Since P is invertible, a linear relation among the columns of A holds if and only if the same relation holds among the columns of R. Therefore, A and R have the same column rank.

In R, the r pivot columns are linearly independent. Every non-pivot column is a linear combination of the pivot columns, as can be seen by solving successively for its entries in the pivot rows. Hence, the column rank of R is also r.

Therefore, both the row rank and column rank of A equal r. ■

Definition 3.4.3 (Rank of a Matrix)**.** The **rank** of a matrix A, denoted by $\operatorname{rank}(A)$, is the common value of its row rank and column rank. Thus, it is the maximum number of linearly independent rows or, equivalently, the maximum number of linearly independent columns.

By the preceding theorem, this definition agrees with the operational definition used in Chapter 2: the rank is the number of non-zero rows in any row echelon form of the matrix.

3.4.2 Determination of Rank by Row Reduction

Example 3.4.4. Find the rank of

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 0 & 5 \end{bmatrix}.$$

Solution. Apply $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -4 \\ 0 & 0 & 3 & -4 \end{bmatrix}.$$

Next, apply $R_3 \to R_3 - R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The resulting matrix has two non-zero rows and hence two pivots. Therefore,

$$\boxed{\operatorname{rank}(A) = 2.}$$

■

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Example 3.4.5. Determine the rank of

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & k \end{bmatrix}$$

for different values of k.

Solution. Applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$, we obtain

$$A \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & k-1 \end{bmatrix}.$$

If $k = 1$, the last row is zero, so the matrix has only one non-zero row and $\operatorname{rank}(A) = 1$.

If $k \neq 1$, the last row is non-zero. Interchanging the last two rows gives a row echelon form with two pivots. Hence, $\operatorname{rank}(A) = 2$.

Therefore,

$$\boxed{\operatorname{rank}(A) = \begin{cases} 1, & k = 1, \\ 2, & k \neq 1. \end{cases}}$$

■

3.5 Rank and the Solution Space of Homogeneous Systems

The rank of a matrix determines the number of pivot variables in the homogeneous system $AX = 0$. The remaining variables are free, and their number determines the dimension of the solution space.

Definition 3.5.1 (Null Space and Nullity)**.** Let $A \in M_{m\times n}(\mathbb{K})$. The **null space** of A, denoted by $N(A)$, is the set of all solutions of the homogeneous system $AX = 0$. Thus,

$$N(A) = \{X \in \mathbb{K}^n : AX = 0\}.$$

The **nullity** of A is the dimension of its null space, denoted by $\operatorname{nullity}(A) = \dim N(A)$.

The null space contains the zero vector. Moreover, if $X, Y \in N(A)$ and $c \in \mathbb{K}$, then $A(X+Y) = AX + AY = 0$ and $A(cX) = cAX = 0$. Hence, the sum of any two solutions and every scalar multiple of a solution are again solutions. Thus, $N(A)$ is a subspace of $\mathbb{K}^n$.

Theorem 3.5.2 (Dimension of the Solution Space)**.** *Let A be an $m \times n$ matrix of rank r. Then*

$$\operatorname{nullity}(A) = n - r.$$

Equivalently,

$$\boxed{\operatorname{rank}(A) + \operatorname{nullity}(A) = n.}$$

Proof. Reduce A to reduced row echelon form. Since $\operatorname{rank}(A) = r$, the system $AX = 0$ has r pivot variables and $n - r$ free variables.

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Let the free variables be $t_1, \ldots, t_{n-r}$. Every pivot variable is a linear combination of these parameters. Consequently, the general solution can be expressed as

$$X = t_1 X_1 + \cdots + t_{n-r} X_{n-r},$$

where X_j is obtained by setting $t_j = 1$ and all the other free parameters equal to zero.

Every solution is a linear combination of $X_1, \ldots, X_{n-r}$, so these vectors span $N(A)$. They are also linearly independent because the free-variable entries of X_j form the standard vectors of $\mathbb{K}^{n-r}$. Thus, $\{X_1, \ldots, X_{n-r}\}$ is a basis of the null space.

Therefore, $\dim N(A) = n - r$, and hence $\text{rank}(A) + \text{nullity}(A) = n$. ■

Example 3.5.3. Let

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 3 \\ 3 & 6 & 2 & 7 \end{bmatrix}.$$

Find the rank and nullity of A. Also find a basis of its null space.

Solution. We first reduce A to reduced row echelon form. Applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$, we obtain

$$A \sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 5 & -5 \\ 0 & 0 & 5 & -5 \end{bmatrix}.$$

Now apply $R_3 \to R_3 - R_2$ and $R_2 \to \frac{1}{5}R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Finally, apply $R_1 \to R_1 + R_2$:

$$\text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The matrix has two pivots, so $\text{rank}(A) = 2$. Since A has four columns,

$$\text{nullity}(A) = 4 - 2 = 2.$$

To find a basis of $N(A)$, solve $AX = 0$. The reduced equations are

$$x_1 + 2x_2 + 3x_4 = 0, \qquad x_3 - x_4 = 0.$$

Thus, $x_1 = -2x_2 - 3x_4$ and $x_3 = x_4$. Let $x_2 = s$ and $x_4 = t$. Then

$$X = \begin{bmatrix} -2s - 3t \\ s \\ t \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Hence, a basis of the null space is

$$\boxed{\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}.}$$

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The basis contains two vectors, confirming that $\text{nullity}(A) = 2$. ■

3.6 Trace of a Matrix and Its Properties

The trace is an important numerical quantity associated with a square matrix. It is particularly easy to calculate, yet it satisfies useful algebraic properties involving matrix addition, scalar multiplication, and matrix products.

Definition 3.6.1 (Trace of a Matrix)**.** Let $A = (a_{ij}) \in M_n(\mathbb{K})$. The **trace** of A, denoted by $\text{Tr}(A)$, is the sum of its main diagonal entries. Thus,

$$\text{Tr}(A) = a_{11} + a_{22} + \cdots + a_{nn} = \sum_{i=1}^{n} a_{ii}.$$

The trace is defined for square matrices.

Example 3.6.2. Find the trace of

$$A = \begin{bmatrix} 3 & -1 & 4 \\ 2 & 5 & 0 \\ 1 & 7 & -2 \end{bmatrix}.$$

Solution. The main diagonal entries are 3, 5, and -2. Therefore,

$$\text{Tr}(A) = 3 + 5 - 2 = \boxed{6}.$$

■

3.6.1 Elementary Properties of Trace

Theorem 3.6.3 (Basic Properties of Trace)**.** *Let $A, B \in M_n(\mathbb{K})$ and $c \in \mathbb{K}$. Then:*

(i) $\text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$.

(ii) $\text{Tr}(cA) = c\,\text{Tr}(A)$.

(iii) $\text{Tr}(A^T) = \text{Tr}(A)$.

(iv) $\text{Tr}(I_n) = n$.

(v) If $A \in M_n(\mathbb{C})$, then $\text{Tr}(A^*) = \overline{\text{Tr}(A)}$.

Proof. Let $A = (a_{ij})$ and $B = (b_{ij})$.

(i) The diagonal entries of $A + B$ are $a_{ii} + b_{ii}$. Therefore,

$$\text{Tr}(A + B) = \sum_{i=1}^{n} (a_{ii} + b_{ii}) = \text{Tr}(A) + \text{Tr}(B).$$

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(ii) The diagonal entries of cA are ca_{ii}. Hence,

$$\operatorname{Tr}(cA) = \sum_{i=1}^{n} ca_{ii} = c\operatorname{Tr}(A).$$

(iii) Transposition does not change the diagonal entries of a matrix. Therefore, $\operatorname{Tr}(A^T) = \operatorname{Tr}(A)$.

(iv) Every diagonal entry of I_n is 1, so $\operatorname{Tr}(I_n) = n$.

(v) The diagonal entries of A^* are $\bar{a}_{ii}$. Hence,

$$\operatorname{Tr}(A^*) = \sum_{i=1}^{n} \bar{a}_{ii} = \overline{\sum_{i=1}^{n} a_{ii}} = \overline{\operatorname{Tr}(A)}.$$

■

Properties (i) and (ii) are called the **linearity properties** of trace. Together, they imply

$$\operatorname{Tr}(cA + dB) = c\operatorname{Tr}(A) + d\operatorname{Tr}(B)$$

for any scalars $c, d \in \mathbb{K}$.

3.6.2 Trace of a Product

Although matrix multiplication is generally non-commutative, the trace of a product satisfies an important symmetry property.

Theorem 3.6.4 (Trace of a Product). *Let $A \in M_{m\times n}(\mathbb{K})$ and $B \in M_{n\times m}(\mathbb{K})$. Then*

$$\boxed{\operatorname{Tr}(AB) = \operatorname{Tr}(BA).}$$

Proof. Let $A = (a_{ij})$ and $B = (b_{ji})$. The i-th diagonal entry of AB is

$$(AB)_{ii} = \sum_{j=1}^{n} a_{ij}b_{ji}.$$

Therefore,

$$\operatorname{Tr}(AB) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ji}.$$

Similarly, the j-th diagonal entry of BA is

$$(BA)_{jj} = \sum_{i=1}^{m} b_{ji}a_{ij},$$

so

$$\operatorname{Tr}(BA) = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji}a_{ij}.$$

Since multiplication of scalars is commutative and the order of summation in a finite sum may be interchanged, the two expressions are equal. Hence, $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$. ■

The matrices AB and BA need not have the same order. However, both products are square, so their traces are defined.

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Example 3.6.5. Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}.$$

Verify that $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$.

Solution. Multiplying the matrices gives

$$AB = \begin{bmatrix} 2 & -1 \\ 10 & 16 \end{bmatrix}.$$

Hence, $\operatorname{Tr}(AB) = 2 + 16 = 18$.

Similarly,

$$BA = \begin{bmatrix} 5 & 3 & 4 \\ -3 & 1 & -4 \\ 10 & -1 & 12 \end{bmatrix}.$$

Therefore, $\operatorname{Tr}(BA) = 5 + 1 + 12 = 18$.

Thus, $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$, even though AB is of order 2×2 and BA is of order 3×3. ■

Theorem 3.6.6 (Cyclic Property of Trace)**.** *Let A, B, and C be matrices of compatible orders such that the products ABC, BCA, and CAB are defined and square. Then*

$$\operatorname{Tr}(ABC) = \operatorname{Tr}(BCA) = \operatorname{Tr}(CAB).$$

Proof. Using the identity $\operatorname{Tr}(XY) = \operatorname{Tr}(YX)$ for matrices of compatible orders, we obtain

$$\operatorname{Tr}(ABC) = \operatorname{Tr}\big(A(BC)\big) = \operatorname{Tr}\big((BC)A\big) = \operatorname{Tr}(BCA).$$

Similarly,

$$\operatorname{Tr}(BCA) = \operatorname{Tr}\big(B(CA)\big) = \operatorname{Tr}\big((CA)B\big) = \operatorname{Tr}(CAB).$$

Combining these equalities gives

$$\operatorname{Tr}(ABC) = \operatorname{Tr}(BCA) = \operatorname{Tr}(CAB).$$

■

The result shows that factors inside a trace may be rearranged cyclically without changing the trace. However, arbitrary permutations of the factors need not preserve the trace.

Exercise 3.6.7. Exercises.

1. Determine whether $v = \begin{bmatrix} 3 \\ 8 \\ 11 \end{bmatrix}$ can be expressed as a linear combination of $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$. If possible, find the coefficients.

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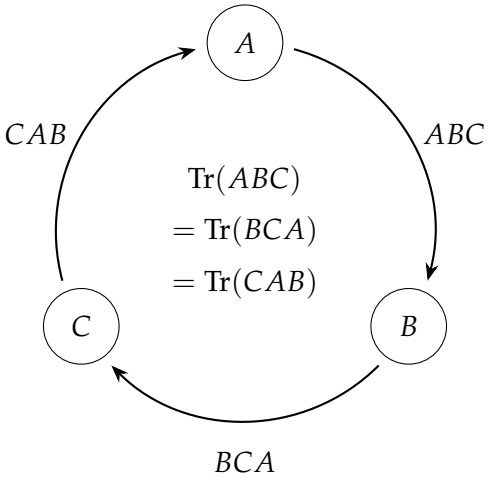


Figure 3.1: Cyclic property of trace. Moving clockwise from any factor preserves the cyclic order and hence the trace.

2. Show that the vectors $\begin{bmatrix}1\\1\\0\end{bmatrix}$, $\begin{bmatrix}0\\1\\1\end{bmatrix}$, and $\begin{bmatrix}1\\0\\1\end{bmatrix}$ span $\mathbb{R}^3$.

3. Show that $\mathcal{B} = \left\{\begin{bmatrix}1+i\\0\end{bmatrix}, \begin{bmatrix}i\\1\end{bmatrix}\right\}$ is a basis of $\mathbb{C}^2$ over $\mathbb{C}$.

4. Determine the value of a for which $v = \begin{bmatrix}1\\a\\3\end{bmatrix}$ is a linear combination of $v_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}$ and $v_2 = \begin{bmatrix}0\\1\\2\end{bmatrix}$.

5. Determine whether the vectors $\begin{bmatrix}1\\2\\3\end{bmatrix}$, $\begin{bmatrix}2\\5\\7\end{bmatrix}$, and $\begin{bmatrix}1\\3\\4\end{bmatrix}$ are linearly independent. If they are dependent, find a non-trivial linear relation among them.

6. Using row reduction, determine whether
$$\begin{bmatrix}1\\0\\1\\2\end{bmatrix}, \quad \begin{bmatrix}0\\1\\-1\\1\end{bmatrix}, \quad \begin{bmatrix}1\\1\\0\\3\end{bmatrix}$$
are linearly independent.

7. Prove that any collection of more than n vectors in $\mathbb{K}^n$ is linearly dependent.

8. Determine all real values of k for which
$$\begin{bmatrix}1\\1\\k\end{bmatrix}, \quad \begin{bmatrix}1\\k\\1\end{bmatrix}, \quad \begin{bmatrix}k\\1\\1\end{bmatrix}$$

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are linearly independent.

9. Show that the vectors $\begin{bmatrix}1\\2\\1\end{bmatrix}$, $\begin{bmatrix}0\\1\\1\end{bmatrix}$, and $\begin{bmatrix}1\\3\\2\end{bmatrix}$ are linearly dependent.

10. Reduce
$$A = \begin{bmatrix}1 & 2 & -1 & 3\\2 & 4 & 1 & 2\\3 & 6 & 0 & 5\end{bmatrix}$$
to row echelon form and determine its rank.

11. Determine the rank of
$$A = \begin{bmatrix}1 & 1 & 2 & 4\\2 & 3 & 7 & 11\\1 & 3 & 8 & 10\end{bmatrix}$$
by elementary row operations.

12. Prove that elementary row operations preserve the row space and hence preserve the rank of a matrix.

13. Prove that $\operatorname{rank}(A) \le \min\{m, n\}$ for every $A \in M_{m\times n}(\mathbb{K})$.

14. Determine the rank of
$$A = \begin{bmatrix}1 & 2 & 1\\2 & 4 & 2\\1 & 2 & k\end{bmatrix}$$
for all values of k.

15. For
$$A = \begin{bmatrix}1 & 2 & -1 & 4\\2 & 4 & 3 & 3\\3 & 6 & 2 & 7\end{bmatrix},$$
find its rank and nullity, the general solution of $AX = 0$, and a basis of its null space.

16. State and prove the rank–nullity theorem. If A is a 4×6 matrix of rank 3, determine its nullity.

17. Find the general solution of $AX = 0$ and a basis of its null space, where
$$A = \begin{bmatrix}1 & -1 & 2 & 3\\2 & 2 & 0 & 2\\4 & 0 & 4 & 8\end{bmatrix}.$$

18. Determine the values of λ for which
$$x + y + z = 0, \qquad x + \lambda y + z = 0, \qquad x + y + \lambda z = 0$$
has a non-trivial solution. Find the complete solution set for these values.

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19. Compute the trace of
$$A = \begin{bmatrix} 3 & -1 & 4 \\ 2 & 5 & -2 \\ 1 & 0 & 6 \end{bmatrix}$$
and verify that $\mathrm{Tr}(A^T) = \mathrm{Tr}(A)$.

20. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ -1 & 5 \end{bmatrix}$. Compute $\mathrm{Tr}(3A - 2B)$ directly and verify the linearity property.

21. For
$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix},$$
compute AB and BA and verify that $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.

CHAPTER

4

MATRIX POLYNOMIALS, EIGENVALUES AND INNER PRODUCTS

In earlier chapters, we studied matrix operations, systems of linear equations, linear independence, and rank. We now use these ideas to study polynomials in square matrices and to determine eigenvalues and eigenvectors. The Cayley–Hamilton theorem provides a useful method for simplifying matrix powers and finding inverses. We then introduce inner products, norms, orthogonality, and orthonormal sets in $\mathbb{R}^n$ and $\mathbb{C}^n$, together with their geometric interpretation.

Throughout this chapter, $\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

4.1 Matrix Polynomials

A polynomial in a scalar variable can also be evaluated at a square matrix. The principal difference is that the constant term must be represented by a scalar multiple of the identity matrix.

Definition 4.1.1 (Polynomial in a Matrix)**.** Let $p(t) = a_0 + a_1 t + \cdots + a_k t^k$ be a polynomial with coefficients in $\mathbb{K}$. For a square matrix $A \in M_n(\mathbb{K})$, the **matrix polynomial** $p(A)$ is defined by

$$p(A) = a_0 I_n + a_1 A + \cdots + a_k A^k,$$

where $A^0 = I_n$.

The constant term a_0 is replaced by $a_0 I_n$ because only matrices of the same order can be added. Notice also that matrix polynomials are defined here for square matrices, since powers such as A^2 and A^3 require matrix multiplication.

Example 4.1.2. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Evaluate $p(A)$ for $p(t) = t^2 - 4t + 3$.

Solution. By definition, $p(A) = A^2 - 4A + 3I_2$. First,

$$A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}.$$

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Therefore,

$$p(A) = \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} - 4\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 1-4+3 & 8-8 \\ 0 & 9-12+3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus, $p(A) = 0$, or equivalently, $A^2 - 4A + 3I_2 = 0$. ■

Definition 4.1.3 (Annihilating Polynomial)**.** A non-zero polynomial $p(t) \in \mathbb{K}[t]$ is called an **annihilating polynomial** of a square matrix A if $p(A) = 0$.

In the preceding example, $t^2 - 4t + 3$ is an annihilating polynomial of A.

Theorem 4.1.4 (Elementary Properties of Matrix Polynomials)**.** *Let $p(t), q(t) \in \mathbb{K}[t]$, $c \in \mathbb{K}$, and $A \in M_n(\mathbb{K})$. Then*

(i) $(p+q)(A) = p(A) + q(A)$;

(ii) $(cp)(A) = c\,p(A)$;

(iii) $(pq)(A) = p(A)q(A)$;

(iv) $p(A)q(A) = q(A)p(A)$.

Proof. The first two identities follow directly from addition and scalar multiplication of matrices.

For the third identity, write $p(t) = \sum_i a_i t^i$ and $q(t) = \sum_j b_j t^j$. Since $A^i A^j = A^{i+j}$, we obtain

$$p(A)q(A) = \sum_{i,j} a_i b_j A^{i+j} = (pq)(A).$$

Finally, since multiplication of scalar polynomials is commutative, $pq = qp$. Hence,

$$p(A)q(A) = (pq)(A) = (qp)(A) = q(A)p(A).$$

■

Thus, polynomials in the same matrix commute with one another, even though multiplication of arbitrary square matrices is generally not commutative.

4.2 Characteristic Polynomial, Eigenvalues and Eigenvectors

The characteristic polynomial of a square matrix provides a systematic method for finding its eigenvalues. These eigenvalues are special scalars associated with vectors whose direction is preserved under multiplication by the matrix.

Definition 4.2.1 (Characteristic Polynomial)**.** For $A \in M_n(\mathbb{K})$, the **characteristic polynomial** of A is

$$p_A(t) = \det(tI_n - A).$$

The equation $p_A(t) = 0$ is called the **characteristic equation** of A.

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The characteristic polynomial has degree n. Its leading coefficient is 1, the coefficient of t^{n-1} is $-\operatorname{Tr}(A)$, and its constant term is $(-1)^n \det(A)$.

For a matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ of order 2,

$$p_A(t) = \begin{vmatrix} t-a & -b \\ -c & t-d \end{vmatrix} = t^2 - (a+d)t + (ad - bc).$$

Thus, for a 2×2 matrix, the characteristic polynomial may be written as $p_A(t) = t^2 - \operatorname{Tr}(A)t + \det(A)$.

4.2.1 Eigenvalues and Eigenvectors

Definition 4.2.2 (Eigenvalue and Eigenvector)**.** Let $A \in M_n(\mathbb{K})$. A scalar $\lambda \in \mathbb{K}$ is called an **eigenvalue** of A if there exists a non-zero vector $v \in \mathbb{K}^n$ satisfying

$$Av = \lambda v.$$

Such a vector v is called an **eigenvector** of A corresponding to λ.

The requirement $v \neq 0$ is essential because the zero vector satisfies $A0 = \lambda 0$ for every scalar λ and therefore cannot be used to distinguish eigenvalues.

Geometrically, for a real matrix, an eigenvector is a non-zero vector whose image under the matrix lies along the same line. The corresponding eigenvalue describes the scaling along that line.

Theorem 4.2.3 (Characteristic Equation and Eigenvalues)**.** *A scalar $\lambda \in \mathbb{K}$ is an eigenvalue of A if and only if*

$$p_A(\lambda) = \det(\lambda I_n - A) = 0.$$

Proof. By definition, λ is an eigenvalue of A if and only if there exists a non-zero vector v such that $Av = \lambda v$. This is equivalent to

$$(\lambda I_n - A)v = 0.$$

The homogeneous system has a non-trivial solution if and only if $\lambda I_n - A$ is singular. This happens precisely when $\det(\lambda I_n - A) = 0$. Hence, the result follows. ■

To find eigenvalues and eigenvectors, first solve the characteristic equation $\det(\lambda I_n - A) = 0$. Then, for each eigenvalue λ, solve $(A - \lambda I_n)v = 0$ to obtain the corresponding non-zero eigenvectors.

Example 4.2.4. Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Solution. The characteristic polynomial is

$$p_A(t) = \begin{vmatrix} t-2 & -1 \\ -1 & t-2 \end{vmatrix} = (t-2)^2 - 1 = (t-1)(t-3).$$

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Thus, the eigenvalues are $\lambda = 1$ and $\lambda = 3$.

For $\lambda = 1$, solve $(A - I_2)v = 0$. Writing $v = \begin{bmatrix} x \\ y \end{bmatrix}$, we obtain

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0.$$

Hence, $x + y = 0$, so $y = -x$. Taking $x = t$, the general solution is

$$v = t \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Therefore, the eigenvectors corresponding to $\lambda = 1$ are all non-zero scalar multiples of $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

For $\lambda = 3$, solve $(A - 3I_2)v = 0$:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0.$$

This gives $x = y$. Hence, the corresponding eigenvectors are all non-zero scalar multiples of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. ■

Example 4.2.5. Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Solution. The characteristic polynomial is

$$\begin{aligned} p_A(t) &= \begin{vmatrix} t-2 & -1 & -1 \\ -1 & t-2 & -1 \\ -1 & -1 & t-2 \end{vmatrix} \\ &= t^3 - 6t^2 + 9t - 4 \\ &= (t-4)(t-1)^2. \end{aligned}$$

Hence, the eigenvalues are $\lambda = 4$ and $\lambda = 1$, where $\lambda = 1$ is repeated twice.

For $\lambda = 4$, we solve $(A - 4I_3)v = 0$:

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

The equations give $x = y = z$. Hence, the corresponding eigenvectors are

$$\boxed{v = s \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad s \neq 0.}$$

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For $\lambda = 1$, we have

$$A - I_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Thus, $(A - I_3)v = 0$ reduces to

$$x + y + z = 0.$$

Taking $y = s$ and $z = t$, we obtain $x = -s - t$. Therefore,

$$\boxed{v = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \qquad (s,t) \neq (0,0).}$$

Thus, the repeated eigenvalue $\lambda = 1$ has two linearly independent eigenvectors. Together with the eigenvector corresponding to $\lambda = 4$, the matrix has three linearly independent eigenvectors. ■

Example 4.2.6. Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 9 & -5 & 5 \\ 7 & -3 & 5 \\ -3 & 3 & -1 \end{bmatrix}.$$

Solution. The characteristic polynomial is

$$\begin{aligned} p_A(t) &= \begin{vmatrix} t-9 & 5 & -5 \\ -7 & t+3 & -5 \\ 3 & -3 & t+1 \end{vmatrix} \\ &= t^3 - 5t^2 + 2t + 8 \\ &= (t+1)(t-2)(t-4). \end{aligned}$$

Hence, the eigenvalues are

$$\lambda_1 = -1, \qquad \lambda_2 = 2, \qquad \lambda_3 = 4.$$

For $\lambda_1 = -1$, we solve $(A + I_3)v = 0$:

$$\begin{bmatrix} 10 & -5 & 5 \\ 7 & -2 & 5 \\ -3 & 3 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

The third equation gives $y = x$. Substituting into the first gives $z = -x$. Thus,

$$\boxed{v = s \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \qquad s \neq 0.}$$

For $\lambda_2 = 2$, we solve $(A - 2I_3)v = 0$:

$$\begin{bmatrix} 7 & -5 & 5 \\ 7 & -5 & 5 \\ -3 & 3 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

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The equations give $x = 0$ and $y = z$. Hence,

$$\boxed{v = s \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad s \neq 0.}$$

For $\lambda_3 = 4$, we solve $(A - 4I_3)v = 0$:

$$\begin{bmatrix} 5 & -5 & 5 \\ 7 & -7 & 5 \\ -3 & 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

Subtracting suitable multiples of the first two equations gives $z = 0$. The first equation then gives $x = y$. Therefore,

$$\boxed{v = s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad s \neq 0.}$$

Thus, A has three distinct real eigenvalues, each with a corresponding linearly independent eigenvector. ■

Example 4.2.7. Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -2 & 2 & 2 \\ -3 & 1 & 5 \end{bmatrix}.$$

Solution. The characteristic polynomial is

$$\begin{aligned} p_A(t) &= \begin{vmatrix} t-1 & -1 & -1 \\ 2 & t-2 & -2 \\ 3 & -1 & t-5 \end{vmatrix} \\ &= t^3 - 8t^2 + 20t - 16 \\ &= (t-2)^2(t-4). \end{aligned}$$

Hence, $\lambda = 2$ is a repeated eigenvalue, while $\lambda = 4$ is a simple eigenvalue.

For $\lambda = 2$, we solve $(A - 2I_3)v = 0$:

$$\begin{bmatrix} -1 & 1 & 1 \\ -2 & 0 & 2 \\ -3 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

The corresponding equations are

$$-x + y + z = 0, \qquad -2x + 2z = 0, \qquad -3x + y + 3z = 0.$$

The second equation gives $z = x$. Substituting into the first gives $y = 0$. The third equation is then automatically satisfied.

Therefore, the corresponding eigenvectors are

$$\boxed{v = s \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad s \neq 0.}$$

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For $\lambda = 4$, we solve $(A - 4I_3)v = 0$:

$$\begin{bmatrix} -3 & 1 & 1 \\ -2 & -2 & 2 \\ -3 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

The equations reduce to

$$-3x + y + z = 0, \qquad -x - y + z = 0.$$

From the second equation, $z = x + y$. Substituting into the first gives $y = x$. Consequently, $z = 2x$.

Hence, the corresponding eigenvectors are

$$\boxed{v = s \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \qquad s \neq 0.}$$

Thus, although $\lambda = 2$ occurs twice in the characteristic polynomial, all its eigenvectors are scalar multiples of a single vector. Consequently, the matrix has only two linearly independent eigenvectors. ■

4.3 Minimal Polynomial of a Matrix

A square matrix may satisfy several polynomial equations. Among all non-zero monic polynomials that annihilate the matrix, the polynomial of least degree is particularly important.

Definition 4.3.1 (Minimal Polynomial)**.** The **minimal polynomial** of a square matrix A, denoted by $m_A(t)$, is the monic polynomial of least positive degree satisfying

$$m_A(A) = 0.$$

A polynomial is called monic if its leading coefficient is 1. The minimal polynomial is unique. Its existence follows from the Cayley–Hamilton theorem, which will be established later in this chapter.

Theorem 4.3.2 (Divisibility Property of the Minimal Polynomial)**.** *Let $m_A(t)$ be the minimal polynomial of A. Then a polynomial $f(t)$ satisfies $f(A) = 0$ if and only if $m_A(t)$ divides $f(t)$.*

Proof. If $m_A(t)$ divides $f(t)$, then $f(t) = m_A(t)q(t)$ for some polynomial $q(t)$. Evaluating at A gives $f(A) = m_A(A)q(A) = 0$.

Conversely, suppose $f(A) = 0$. By the polynomial division algorithm, write

$$f(t) = q(t)m_A(t) + r(t),$$

where $r(t) = 0$ or $\deg r < \deg m_A$. Substituting A gives $r(A) = 0$. If r were non-zero, it would be an annihilating polynomial of smaller degree than m_A, contradicting minimality. Therefore, $r = 0$, and hence m_A divides f. ■

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Example 4.3.3. Find the minimal polynomial of

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Solution. The characteristic polynomial is

$$\begin{aligned} p_A(t) &= \begin{vmatrix} t-2 & -1 & -1 \\ -1 & t-2 & -1 \\ -1 & -1 & t-2 \end{vmatrix} \\ &= t^3 - 6t^2 + 9t - 4 \\ &= (t-1)^2(t-4). \end{aligned}$$

Hence, the minimal polynomial divides $(t-1)^2(t-4)$.

Since 1 and 4 are distinct eigenvalues, both $t-1$ and $t-4$ must be factors of $m_A(t)$. The smallest possible minimal polynomial is therefore

$$q(t) = (t-1)(t-4) = t^2 - 5t + 4.$$

We compute

$$A^2 = \begin{bmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{bmatrix}.$$

Consequently,

$$\begin{aligned} q(A) &= A^2 - 5A + 4I_3 \\ &= \begin{bmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{bmatrix} - \begin{bmatrix} 10 & 5 & 5 \\ 5 & 10 & 5 \\ 5 & 5 & 10 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \\ &= 0. \end{aligned}$$

Thus, $q(t)$ annihilates A. Since it contains both distinct eigenvalue factors, no polynomial of smaller degree can annihilate A. Therefore,

$$\boxed{m_A(t) = (t-1)(t-4).}$$

■

Example 4.3.4. Find the minimal polynomial of

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution. The characteristic polynomial is

$$\begin{aligned} p_A(t) &= \begin{vmatrix} t-1 & -1 & -1 \\ -1 & t-1 & 0 \\ -1 & 0 & t-1 \end{vmatrix} \\ &= (t-1)\big((t-1)^2 - 2\big) \\ &= (t-1)(t-1-\sqrt{2})(t-1+\sqrt{2}). \end{aligned}$$

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Thus, A has three distinct eigenvalues:

$$1, \qquad 1+\sqrt{2}, \qquad 1-\sqrt{2}.$$

Every eigenvalue must be a root of the minimal polynomial. Hence, $m_A(t)$ must contain the three distinct linear factors of $p_A(t)$.

Since the characteristic polynomial already annihilates A by the Cayley–Hamilton theorem, it follows that

$$\boxed{m_A(t) = (t-1)(t^2-2t-1).}$$

Thus, in this example, the minimal polynomial is equal to the characteristic polynomial. ■

Example 4.3.5. Find the minimal polynomial of

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -2 & 2 & 2 \\ -3 & 1 & 5 \end{bmatrix}.$$

Solution. The characteristic polynomial is

$$\begin{aligned} p_A(t) &= \begin{vmatrix} t-1 & -1 & -1 \\ 2 & t-2 & -2 \\ 3 & -1 & t-5 \end{vmatrix} \\ &= t^3 - 8t^2 + 20t - 16 \\ &= (t-2)^2(t-4). \end{aligned}$$

The distinct eigenvalues are 2 and 4. Therefore, the minimal polynomial must contain both factors $t-2$ and $t-4$.

The first candidate is

$$q(t) = (t-2)(t-4) = t^2 - 6t + 8.$$

We calculate

$$A - 2I_3 = \begin{bmatrix} -1 & 1 & 1 \\ -2 & 0 & 2 \\ -3 & 1 & 3 \end{bmatrix}, \qquad A - 4I_3 = \begin{bmatrix} -3 & 1 & 1 \\ -2 & -2 & 2 \\ -3 & 1 & 1 \end{bmatrix}.$$

Multiplying these matrices gives

$$q(A) = (A-2I_3)(A-4I_3) = \begin{bmatrix} -2 & -2 & 2 \\ 0 & 0 & 0 \\ -2 & -2 & 2 \end{bmatrix} \neq 0.$$

Thus, $q(t)$ does not annihilate A.

The only remaining monic divisor of the characteristic polynomial containing both distinct factors is

$$(t-2)^2(t-4).$$

By the Cayley–Hamilton theorem, this polynomial annihilates A. Therefore,

$$\boxed{m_A(t) = (t-2)^2(t-4).}$$

In this example, the factor corresponding to the repeated eigenvalue 2 must occur twice in the minimal polynomial. ■

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Example 4.3.6. Find the minimal polynomial of

$$A = \begin{bmatrix} 1 & 1 & -2 \\ 1 & 1 & -2 \\ 1 & 1 & -2 \end{bmatrix}.$$

Solution. The characteristic polynomial is

$$p_A(t) = \begin{vmatrix} t-1 & -1 & 2 \\ -1 & t-1 & 2 \\ -1 & -1 & t+2 \end{vmatrix} = t^3.$$

Thus, the only eigenvalue is 0, and the minimal polynomial must be one of

$$t, \qquad t^2, \qquad t^3.$$

Clearly, $A \neq 0$. However, direct multiplication gives

$$A^2 = \begin{bmatrix} 1 & 1 & -2 \\ 1 & 1 & -2 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 & -2 \\ 1 & 1 & -2 \\ 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore, t^2 annihilates A, whereas t does not. Hence,

$$\boxed{m_A(t) = t^2.}$$

Notice that the characteristic polynomial is t^3, while the minimal polynomial is t^2. ■

4.3.1 Minimal Polynomials of Some Special Matrices

The minimal polynomials of idempotent, involutory, and nilpotent matrices can be determined directly from their defining equations.

Theorem 4.3.7 (Minimal Polynomial of an Idempotent Matrix)**.** *Let $A \in M_n(\mathbb{K})$ be idempotent, that is, $A^2 = A$. Then*

$$m_A(t) = \begin{cases} t, & A = 0, \\ t-1, & A = I_n, \\ t(t-1), & A \neq 0,\ A \neq I_n. \end{cases}$$

Proof. Since $A^2 = A$, we have

$$A^2 - A = 0.$$

Thus, the polynomial $t^2 - t = t(t-1)$ annihilates A. By the divisibility property,

$$m_A(t) \mid t(t-1).$$

The possible monic divisors of positive degree are t, $t-1$, and $t(t-1)$.
If $A = 0$, then t annihilates A, giving $m_A(t) = t$.
If $A = I_n$, then $A - I_n = 0$, giving $m_A(t) = t - 1$.
Finally, if $A \neq 0$ and $A \neq I_n$, neither t nor $t-1$ annihilates A. Hence, the minimal polynomial must be $t(t-1)$. ■

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Example 4.3.8. Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since $A^2 = A$ and A is neither the zero matrix nor the identity matrix, its minimal polynomial is

$$\boxed{m_A(t) = t(t-1).}$$

Theorem 4.3.9 (Minimal Polynomial of an Involutory Matrix). *Let $A \in M_n(\mathbb{K})$ be involutory, that is, $A^2 = I_n$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then*

$$m_A(t) = \begin{cases} t-1, & A = I_n, \\ t+1, & A = -I_n, \\ (t-1)(t+1), & A \neq I_n,\ A \neq -I_n. \end{cases}$$

Proof. Since $A^2 = I_n$, we obtain

$$A^2 - I_n = 0.$$

Therefore, $(t-1)(t+1)$ annihilates A, and hence

$$m_A(t) \mid (t-1)(t+1).$$

Over $\mathbb{R}$ or $\mathbb{C}$, the factors $t-1$ and $t+1$ are distinct. Thus, the monic divisors of positive degree are $t-1$, $t+1$, and $(t-1)(t+1)$.

If $A = I_n$, then $m_A(t) = t-1$. If $A = -I_n$, then $m_A(t) = t+1$. Otherwise, neither linear factor annihilates A, so

$$m_A(t) = (t-1)(t+1).$$

■

Example 4.3.10. Let

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

Since $A^2 = I_3$, but $A \neq I_3$ and $A \neq -I_3$, we have

$$\boxed{m_A(t) = t^2 - 1.}$$

Theorem 4.3.11 (Minimal Polynomial of a Nilpotent Matrix). *Let $A \in M_n(\mathbb{K})$ be nilpotent of index k, that is,*

$$A^k = 0, \qquad A^{k-1} \neq 0,$$

where k is the smallest positive integer for which $A^k = 0$. Then

$$\boxed{m_A(t) = t^k.}$$

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Proof. Since $A^k = 0$, the polynomial t^k annihilates A. Therefore,

$$m_A(t) \mid t^k.$$

Every monic divisor of t^k is of the form t^r for some integer $0 \leq r \leq k$.

Since the minimal polynomial has positive degree, $r \geq 1$. If $r < k$, then $A^r = 0$, contradicting the definition of the index of nilpotency.

Consequently, $r = k$, and hence $m_A(t) = t^k$. ■

Example 4.3.12. Let

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

We have

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq 0, \qquad A^3 = 0.$$

Thus, A is nilpotent of index 3, and its minimal polynomial is

$$\boxed{m_A(t) = t^3.}$$

4.4 Eigenvalues and Eigenvectors under Similarity

A linear transformation can be represented by different matrices when different bases are chosen. The matrices may have different entries, but they possess the same eigenvalues.

Definition 4.4.1 (Similar Matrices)**.** Two square matrices $A, B \in M_n(\mathbb{K})$ are called **similar** if there exists a non-singular matrix P such that

$$B = P^{-1}AP.$$

Theorem 4.4.2 (Invariance of Eigenvalues under Similarity)**.** *If A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues. Moreover, if $Av = \lambda v$ with $v \neq 0$, then $P^{-1}v$ is an eigenvector of B corresponding to the same eigenvalue λ, where $B = P^{-1}AP$.*

Proof. Since $B = P^{-1}AP$, we have $\lambda I_n - B = P^{-1}(\lambda I_n - A)P$. Taking determinants gives

$$\begin{aligned} \det(\lambda I_n - B) &= \det(P^{-1}) \det(\lambda I_n - A) \det(P) \\ &= \det(\lambda I_n - A). \end{aligned}$$

Therefore, A and B have the same characteristic polynomial and the same eigenvalues.

Now suppose $Av = \lambda v$ with $v \neq 0$. Then

$$B(P^{-1}v) = P^{-1}APP^{-1}v = P^{-1}Av = \lambda P^{-1}v.$$

Since P is invertible, $P^{-1}v \neq 0$. Hence, $P^{-1}v$ is an eigenvector of B corresponding to λ. ■

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Example 4.4.3. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Verify that $B = P^{-1}AP$ has the same eigenvalues as A.

Solution. We have $P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Thus,

$$B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}.$$

Since A and B are triangular matrices with diagonal entries 2 and 3, both have characteristic polynomial $(t-2)(t-3)$. Hence, both have eigenvalues 2 and 3. ■

4.5 The Cayley–Hamilton Theorem

The Cayley–Hamilton theorem establishes a fundamental relationship between a square matrix and its characteristic polynomial. It is particularly useful for reducing higher powers of matrices and finding inverses.

Theorem 4.5.1 (Cayley–Hamilton Theorem)**.** *Every square matrix satisfies its own characteristic equation. More precisely, if $A \in M_n(\mathbb{K})$ has characteristic polynomial*

$$p_A(t) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0,$$

then

$$p_A(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0.$$

Proof. Let $A \in M_n(\mathbb{K})$, and consider its characteristic matrix $tI_n - A$. By the fundamental identity for the adjugate of a square matrix,

$$(tI_n - A)\operatorname{adj}(tI_n - A) = \det(tI_n - A)I_n = p_A(t)I_n.$$

Put $D(t) = \operatorname{adj}(tI_n - A)$. Since each entry of $D(t)$ is a cofactor of $tI_n - A$, it is a polynomial in t of degree at most $n-1$. Therefore, we can express $D(t)$ as

$$D(t) = D_0 + D_1 t + D_2 t^2 + \cdots + D_{n-1}t^{n-1},$$

where $D_0, D_1, \ldots, D_{n-1}$ are constant square matrices of order n.

Substituting this expression into the adjugate identity, we obtain

$$(tI_n - A)(D_0 + D_1 t + \cdots + D_{n-1}t^{n-1}) = (t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0)I_n.$$

Expanding the left-hand side and collecting like powers of t gives

$$\begin{aligned} &-AD_0 + (D_0 - AD_1)t + (D_1 - AD_2)t^2 \\ &\quad + \cdots + (D_{n-2} - AD_{n-1})t^{n-1} + D_{n-1}t^n \\ &= c_0 I_n + c_1 t I_n + \cdots + c_{n-1}t^{n-1}I_n + t^n I_n. \end{aligned}$$

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Since these two matrix polynomials are equal, the coefficients of corresponding powers of t must be equal. Hence,

$$\begin{aligned} -AD_0 &= c_0 I_n, \\ D_0 - AD_1 &= c_1 I_n, \\ D_1 - AD_2 &= c_2 I_n, \\ &\vdots \\ D_{n-2} - AD_{n-1} &= c_{n-1} I_n, \\ D_{n-1} &= I_n. \end{aligned}$$

We now multiply these equations successively on the left by $I_n, A, A^2, \ldots, A^{n-1}, A^n$, respectively. This gives

$$\begin{aligned} -AD_0 &= c_0 I_n, \\ AD_0 - A^2 D_1 &= c_1 A, \\ A^2 D_1 - A^3 D_2 &= c_2 A^2, \\ &\vdots \\ A^{n-1} D_{n-2} - A^n D_{n-1} &= c_{n-1} A^{n-1}, \\ A^n D_{n-1} &= A^n. \end{aligned}$$

Adding all these equations, we observe that the terms involving the matrices $D_0, D_1, \ldots, D_{n-1}$ cancel in pairs:

$$\begin{aligned} &(-AD_0) + (AD_0 - A^2 D_1) + (A^2 D_1 - A^3 D_2) \\ &\quad + \cdots + (A^{n-1} D_{n-2} - A^n D_{n-1}) + A^n D_{n-1} \\ &= c_0 I_n + c_1 A + c_2 A^2 + \cdots + c_{n-1} A^{n-1} + A^n. \end{aligned}$$

The entire left-hand side is zero. Consequently,

$$A^n + c_{n-1} A^{n-1} + \cdots + c_1 A + c_0 I_n = 0.$$

The expression on the left is precisely $p_A(A)$. Therefore,

$$\boxed{p_A(A) = 0.}$$

Thus, every square matrix satisfies its own characteristic equation. ■

Corollary 4.5.2. *The minimal polynomial of a square matrix divides its characteristic polynomial.*

Proof. By the Cayley–Hamilton theorem, $p_A(A) = 0$. The divisibility property of the minimal polynomial therefore gives $m_A(t) \mid p_A(t)$. ■

4.5.1 Applications of the Cayley–Hamilton Theorem

The Cayley–Hamilton theorem is particularly useful in matrix computations. It enables us to express higher powers of a square matrix in terms of lower powers and to determine the inverse of a non-singular matrix without using the adjugate method.

If the characteristic polynomial of $A \in M_n(\mathbb{K})$ is

$$p_A(t) = t^n + c_{n-1} t^{n-1} + \cdots + c_1 t + c_0,$$

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then, by the Cayley–Hamilton theorem,

$$A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0.$$

Consequently, A^n can be expressed as a linear combination of $I_n, A, \ldots, A^{n-1}$. Multiplying this relation by successive powers of A gives corresponding expressions for higher powers of A.

If A is non-singular, then $c_0 = (-1)^n \det(A) \neq 0$. Rearranging the characteristic equation gives

$$A\left(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I_n\right) = -c_0 I_n.$$

Hence,

$$\boxed{A^{-1} = -\frac{1}{c_0}\left(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I_n\right).}$$

We illustrate these applications through the following examples.

Example 4.5.3. Verify the Cayley–Hamilton theorem for

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and hence find A^{-1}.

Solution. The characteristic polynomial is

$$\begin{aligned} p_A(t) &= \begin{vmatrix} t-1 & -2 \\ -3 & t-4 \end{vmatrix} \\ &= (t-1)(t-4) - 6 = t^2 - 5t - 2. \end{aligned}$$

By the Cayley–Hamilton theorem, $A^2 - 5A - 2I_2 = 0$. To verify this directly, we calculate

$$A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} A^2 - 5A - 2I_2 &= \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Thus, the theorem is verified.

To find A^{-1}, rearrange the characteristic equation:

$$A^2 - 5A = 2I_2 \quad \Longrightarrow \quad A(A - 5I_2) = 2I_2.$$

Hence,

$$\boxed{A^{-1} = \frac{1}{2}(A - 5I_2) = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}.}$$

■

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Example 4.5.4. For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, use the Cayley–Hamilton theorem to express A^4 as a linear combination of A and I_2.

Solution. From the characteristic equation, $A^2 = 5A + 2I_2$. Multiplying by A gives

$$\begin{aligned} A^3 &= 5A^2 + 2A \\ &= 5(5A + 2I_2) + 2A = 27A + 10I_2. \end{aligned}$$

Multiplying once more by A, we obtain

$$\begin{aligned} A^4 &= 27A^2 + 10A \\ &= 27(5A + 2I_2) + 10A \\ &= 145A + 54I_2. \end{aligned}$$

Therefore,

$$\boxed{A^4 = 145A + 54I_2 = \begin{bmatrix} 199 & 290 \\ 435 & 634 \end{bmatrix}.}$$

■

Example 4.5.5. Use the Cayley–Hamilton theorem to find the inverse of

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$$

Solution. First, we determine the characteristic polynomial:

$$\begin{aligned} p_A(t) &= \begin{vmatrix} t-2 & -1 & 0 \\ 0 & t-2 & -1 \\ -1 & 0 & t-2 \end{vmatrix} \\ &= (t-2)^3 - 1 \\ &= t^3 - 6t^2 + 12t - 9. \end{aligned}$$

By the Cayley–Hamilton theorem,

$$A^3 - 6A^2 + 12A - 9I_3 = 0.$$

Rearranging and taking A as a common factor, we obtain

$$A(A^2 - 6A + 12I_3) = 9I_3.$$

Since $\det(A) = 9 \neq 0$, the matrix A is non-singular. Consequently,

$$A^{-1} = \frac{1}{9}(A^2 - 6A + 12I_3).$$

We now calculate A^2:

$$A^2 = \begin{bmatrix} 4 & 4 & 1 \\ 1 & 4 & 4 \\ 4 & 1 & 4 \end{bmatrix}.$$

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Therefore,

$$A^2 - 6A + 12I_3 = \begin{bmatrix} 4 & 4 & 1 \\ 1 & 4 & 4 \\ 4 & 1 & 4 \end{bmatrix} - \begin{bmatrix} 12 & 6 & 0 \\ 0 & 12 & 6 \\ 6 & 0 & 12 \end{bmatrix} + \begin{bmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$
$$= \begin{bmatrix} 4 & -2 & 1 \\ 1 & 4 & -2 \\ -2 & 1 & 4 \end{bmatrix}.$$

Hence,

$$\boxed{A^{-1} = \frac{1}{9} \begin{bmatrix} 4 & -2 & 1 \\ 1 & 4 & -2 \\ -2 & 1 & 4 \end{bmatrix}.}$$

■

Example 4.5.6. Use the Cayley–Hamilton theorem to find the inverse of

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Solution. The characteristic polynomial is

$$p_A(t) = \begin{vmatrix} t-2 & -1 & -1 \\ -1 & t-2 & -1 \\ -1 & -1 & t-2 \end{vmatrix}$$
$$= (t-4)(t-1)^2$$
$$= t^3 - 6t^2 + 9t - 4.$$

By the Cayley–Hamilton theorem,

$$A^3 - 6A^2 + 9A - 4I_3 = 0.$$

Rearranging,

$$A(A^2 - 6A + 9I_3) = 4I_3.$$

Since $\det(A) = 4 \neq 0$, we obtain

$$A^{-1} = \frac{1}{4}(A^2 - 6A + 9I_3).$$

Now

$$A^2 = \begin{bmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{bmatrix}.$$

Therefore,

$$A^2 - 6A + 9I_3 = \begin{bmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{bmatrix} - \begin{bmatrix} 12 & 6 & 6 \\ 6 & 12 & 6 \\ 6 & 6 & 12 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$
$$= \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}.$$

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Hence,

$$\boxed{A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}.}$$

■

Example 4.5.7. Use the Cayley–Hamilton theorem to express A^4 as a linear combination of A^2, A, and I_3, where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

Hence, calculate A^4.

Solution. The characteristic polynomial is

$$\begin{aligned} p_A(t) &= \begin{vmatrix} t-1 & -1 & -1 \\ -1 & t-1 & 0 \\ -1 & 0 & t-1 \end{vmatrix} \\ &= (t-1)\big((t-1)^2 - 2\big) \\ &= t^3 - 3t^2 + t + 1. \end{aligned}$$

By the Cayley–Hamilton theorem,

$$A^3 - 3A^2 + A + I_3 = 0.$$

Thus,

$$A^3 = 3A^2 - A - I_3.$$

Multiplying this equality by A, we obtain

$$A^4 = 3A^3 - A^2 - A.$$

Substituting the expression for A^3 gives

$$\begin{aligned} A^4 &= 3(3A^2 - A - I_3) - A^2 - A \\ &= 9A^2 - 3A - 3I_3 - A^2 - A \\ &= 8A^2 - 4A - 3I_3. \end{aligned}$$

Therefore,

$$\boxed{A^4 = 8A^2 - 4A - 3I_3.}$$

To calculate A^4, first find

$$A^2 = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}.$$

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Hence,

$$A^4 = 8\begin{bmatrix}3 & 2 & 2\\ 2 & 2 & 1\\ 2 & 1 & 2\end{bmatrix} - 4\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 0\\ 1 & 0 & 1\end{bmatrix} - 3I_3$$
$$= \boxed{\begin{bmatrix}17 & 12 & 12\\ 12 & 9 & 8\\ 12 & 8 & 9\end{bmatrix}}.$$

■

Example 4.5.8. Use the Cayley–Hamilton theorem to find A^5 for

$$A = \begin{bmatrix}2 & 1 & 1\\ 1 & 2 & 1\\ 1 & 1 & 2\end{bmatrix}.$$

Solution. As obtained earlier, the characteristic polynomial of A is

$$p_A(t) = t^3 - 6t^2 + 9t - 4.$$

Thus, the Cayley–Hamilton theorem gives

$$A^3 = 6A^2 - 9A + 4I_3.$$

Multiplying by A,

$$\begin{aligned} A^4 &= 6A^3 - 9A^2 + 4A \\ &= 6(6A^2 - 9A + 4I_3) - 9A^2 + 4A \\ &= 27A^2 - 50A + 24I_3. \end{aligned}$$

Again multiplying by A, we obtain

$$\begin{aligned} A^5 &= 27A^3 - 50A^2 + 24A \\ &= 27(6A^2 - 9A + 4I_3) - 50A^2 + 24A \\ &= 112A^2 - 219A + 108I_3. \end{aligned}$$

Now

$$A^2 = \begin{bmatrix}6 & 5 & 5\\ 5 & 6 & 5\\ 5 & 5 & 6\end{bmatrix}.$$

Therefore,

$$A^5 = 112\begin{bmatrix}6 & 5 & 5\\ 5 & 6 & 5\\ 5 & 5 & 6\end{bmatrix} - 219\begin{bmatrix}2 & 1 & 1\\ 1 & 2 & 1\\ 1 & 1 & 2\end{bmatrix} + 108I_3$$
$$= \boxed{\begin{bmatrix}342 & 341 & 341\\ 341 & 342 & 341\\ 341 & 341 & 342\end{bmatrix}}.$$

■

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4.6 Inner Products, Norms and Orthogonality

These notes follow the syllabus order. Throughout, $\mathbb{K}$ denotes $\mathbb{R}$ or $\mathbb{C}$, and vectors are column vectors. We adopt the convention that an inner product is linear in its *first* argument.

4.6.1 Inner Products on $\mathbb{R}^n$ and $\mathbb{C}^n$

Definition 4.6.1 (Inner Product)**.** An **inner product** on $\mathbb{K}^n$ is a function $\langle \cdot, \cdot \rangle : \mathbb{K}^n \times \mathbb{K}^n \to \mathbb{K}$ such that, for $x, y, z \in \mathbb{K}^n$ and $a, b \in \mathbb{K}$,

(i) $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$;

(ii) $\langle x, y \rangle = \overline{\langle y, x \rangle}$;

(iii) $\langle x, x \rangle \geq 0$, with equality if and only if $x = 0$.

A vector space with an inner product is called an **inner product space**. Conjugate symmetry gives $\langle x, ay + bz \rangle = \overline{a}\langle x, y \rangle + \overline{b}\langle x, z \rangle$.

Definition 4.6.2 (Standard Inner Products)**.** For $x = (x_1, \ldots, x_n)^T$ and $y = (y_1, \ldots, y_n)^T$, the standard inner products are

$$\langle x, y \rangle = \sum_{j=1}^{n} x_j y_j = x^T y \quad (\mathbb{R}^n), \qquad \langle x, y \rangle = \sum_{j=1}^{n} x_j \overline{y_j} = y^* x \quad (\mathbb{C}^n).$$

Here y^* denotes the conjugate transpose.

Proposition 4.6.3. The standard inner products satisfy the inner product axioms.

Proof. Linearity follows from expansion, conjugate symmetry follows by conjugating the sum, and $\langle x, x \rangle = \sum_{j=1}^{n} |x_j|^2 \geq 0$, with equality precisely for $x = 0$. ∎

Example 4.6.4. Compute $\langle x, y \rangle$ for $x = (1, 2, -3)^T$ and $y = (4, -1, 2)^T$ in $\mathbb{R}^3$.

Solution. $\langle x, y \rangle = 1(4) + 2(-1) + (-3)(2) = -4$. ∎

Example 4.6.5. Compute the standard inner product of $x = (1 + i, 2)^T$ and $y = (i, 1 - i)^T$ in $\mathbb{C}^2$.

Solution. Conjugating the entries of the second vector gives

$$\langle x, y \rangle = (1 + i)(-i) + 2(1 + i) = \boxed{3 + i}.$$

∎

Example 4.6.6. Compute $\langle x, y \rangle$ for $x = (1 + i, 2)^T$ and $y = (3i, 1 - i)^T$ in $\mathbb{C}^2$.

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Solution.

$$\langle x, y\rangle = (1+i)(-3i) + 2(1+i) = \boxed{5 - i}.$$

■

Example 4.6.7. Let $x = \begin{bmatrix} 1+i \\ 2 \\ -i \end{bmatrix}$ and $y = \begin{bmatrix} 2i \\ 1-i \\ 3 \end{bmatrix}$ in $\mathbb{C}^3$. Compute $\langle x, y\rangle$, $\|x\|$, $\|y\|$, and verify $\langle y, x\rangle = \overline{\langle x, y\rangle}$.

Solution. Under the convention $\langle x, y\rangle = \sum_j x_j \overline{y_j}$,

$$\langle x, y\rangle = (1+i)(-2i) + 2(1+i) + (-i)(3) = 4 - 3i,$$
$$\langle y, x\rangle = \overline{\langle x, y\rangle} = 4 + 3i.$$

Also, $\|x\| = \sqrt{2+4+1} = \sqrt{7}$ and $\|y\| = \sqrt{4+2+9} = \sqrt{15}$. Thus $\langle y, x\rangle = \overline{\langle x, y\rangle}$. ■

4.6.2 Norms

Definition 4.6.8 (Norm and Unit Vector)**.** The **norm** of $x \in \mathbb{K}^n$ is $\|x\| = \sqrt{\langle x, x\rangle} = \left(\sum_{j=1}^{n} |x_j|^2\right)^{1/2}$. A vector of norm 1 is called a **unit vector**. For $x \neq 0$, the vector $x/\|x\|$ is the normalization of x.

Example 4.6.9. Find the norm of $x = (2, -3, 6)^T$ and a unit vector in its direction.

Solution. $\|x\| = \sqrt{2^2 + (-3)^2 + 6^2} = 7$, so the required unit vector is $\boxed{\frac{1}{7}(2, -3, 6)^T}$. ■

Example 4.6.10. Find the norm of $z = (1+i, 1-i)^T \in \mathbb{C}^2$.

Solution. $\|z\| = \sqrt{|1+i|^2 + |1-i|^2} = \sqrt{2+2} = 2$. ■

4.6.3 Orthogonality

Definition 4.6.11 (Orthogonal Vectors)**.** Two vectors $x, y \in \mathbb{K}^n$ are **orthogonal** if $\langle x, y\rangle = 0$, written $x \perp y$. A set of non-zero vectors is **orthogonal** if any two distinct vectors in it are orthogonal.

Theorem 4.6.12 (Linear Independence of Orthogonal Vectors)**.** *Every orthogonal set of non-zero vectors is linearly independent.*

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Proof. If $\sum_{j=1}^{k} c_j v_j = 0$ and the v_j are non-zero and mutually orthogonal, take the inner product with v_i in the second argument. Then $c_i \|v_i\|^2 = 0$, so $c_i = 0$ for each i. ■

Example 4.6.13. Show that $u = (1,1,0)^T$, $v = (1,-1,0)^T$, and $w = (0,0,2)^T$ are linearly independent.

Solution. The three pairwise inner products are $\langle u, v\rangle = 1 - 1 = 0$, $\langle u, w\rangle = 0$, and $\langle v, w\rangle = 0$. None of the vectors is zero; therefore the preceding theorem implies linear independence. ■

4.6.4 Orthonormal Sets

Definition 4.6.14 (Orthonormal Set)**.** An orthogonal set $\{u_1, \dots, u_k\}$ is **orthonormal** if $\|u_i\| = 1$ for all i. Equivalently, $\langle u_i, u_j\rangle = \delta_{ij}$, where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise.

Example 4.6.15. Show that $u = \dfrac{1}{\sqrt{2}}(1,1)^T$ and $v = \dfrac{1}{\sqrt{2}}(1,-1)^T$ form an orthonormal set in $\mathbb{R}^2$.

Solution. $\langle u, v\rangle = \dfrac{1}{2}(1-1) = 0$ and $\|u\|^2 = \|v\|^2 = \dfrac{1}{2}(1+1) = 1$. Hence $\{u, v\}$ is orthonormal. ■

4.6.5 Geometric Interpretation of Length and Angles

For $x \in \mathbb{R}^n$, $\|x\|$ is its Euclidean length, while the distance between $x, y \in \mathbb{R}^n$ is $d(x,y) = \|x - y\|$. For non-zero real vectors, the angle $0 \le \theta \le \pi$ is determined by

$$\boxed{\cos\theta = \frac{\langle x, y\rangle}{\|x\|\,\|y\|}}.$$

Thus, two non-zero real vectors are perpendicular exactly when their inner product is zero. For complex vectors, orthogonality is still defined by the inner product, but this real angle formula is not used.

Example 4.6.16. Find the angle between $x = (1,0)^T$ and $y = (1,1)^T$.

Solution. Since $\langle x, y\rangle = 1$, $\|x\| = 1$, and $\|y\| = \sqrt{2}$, we have $\cos\theta = \dfrac{1}{\sqrt{2}}$. Therefore $\boxed{\theta = \dfrac{\pi}{4} = 45^\circ}$. ■

Example 4.6.17. Find the angle between $x = (1,1)^T$ and $y = (-1,1)^T$.

Solution. $\langle x, y\rangle = -1 + 1 = 0$, so $\boxed{\theta = \dfrac{\pi}{2} = 90^\circ}$. ■

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4.7 Additional Recommended Topics

The following results and techniques extend the prescribed syllabus and provide useful preparation for further linear algebra.

4.7.1 Other Inner Products

The standard inner products are not the only possible inner products. The following examples illustrate inner products on vectors, matrices, and function spaces.

Example 4.7.1 (Weighted Inner Product)**.** Let $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. Verify that $\langle x, y \rangle_A = x^T A y$ defines an inner product on $\mathbb{R}^2$, and evaluate it for $x = (1,2)^T$, $y = (4,1)^T$.

Solution. Since A is symmetric with positive diagonal entries, $x^T A x = 2x_1^2 + 3x_2^2 > 0$ for $x \neq 0$; the remaining axioms follow from matrix multiplication. Further,

$$\langle x, y \rangle_A = 2(1)(4) + 3(2)(1) = 14.$$

More generally, any real symmetric positive-definite A defines $\langle x, y \rangle_A = x^T A y$. ■

Example 4.7.2 (Frobenius Inner Product on Real Matrices)**.** For real matrices of the same size, set $\langle A, B \rangle_F = \operatorname{Tr}(AB^T) = \sum_{i,j} a_{ij} b_{ij}$. Find this product for $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ 4 & 1 \end{pmatrix}$.

Solution. $\langle A, B \rangle_F = 1(2) + 2(-1) + 0(4) + 3(1) = 3$. ■

Example 4.7.3 (Frobenius Inner Product on Complex Matrices)**.** For complex matrices of the same size, set $\langle A, B \rangle_F = \operatorname{Tr}(AB^*) = \sum_{i,j} a_{ij} \overline{b_{ij}}$. Evaluate it for $A = \begin{pmatrix} i & 1 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2i & 0 \\ 1 & 1+i \end{pmatrix}$.

Solution.

$$\langle A, B \rangle_F = i(-2i) + 1(0) + 0(1) + 2(1-i) = \boxed{4 - 2i}.$$

■

Example 4.7.4 (Inner Product of Real-Valued Functions)**.** On $C[a,b]$, the space of continuous real-valued functions, the formula $\langle f, g \rangle = \int_a^b f(t) g(t)\, dt$ defines an inner product. Find $\langle f, g \rangle$ on $[0,1]$ for $f(t) = t$ and $g(t) = t^2$.

Solution. $\langle f, g \rangle = \displaystyle\int_0^1 t^3\, dt = \left[\frac{t^4}{4} \right]_0^1 = \boxed{\frac{1}{4}}$. ■

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Example 4.7.5 (Inner Product of Complex-Valued Functions)**.** For continuous complex-valued functions use $\langle f, g\rangle = \int_a^b f(t)\overline{g(t)}\,dt$. Show that $f(t) = e^{it}$ and $g(t) = e^{2it}$ are orthogonal on $[0, 2\pi]$.

Solution.

$$\langle f, g\rangle = \int_0^{2\pi} e^{it}e^{-2it}\,dt = \int_0^{2\pi} e^{-it}\,dt = \left[\frac{e^{-it}}{-i}\right]_0^{2\pi} = 0.$$

■

4.7.2 Cauchy–Schwarz Inequality and Properties of Norms

Theorem 4.7.6 (Cauchy–Schwarz Inequality)**.** *For $x, y \in \mathbb{K}^n$, $|\langle x, y\rangle| \le \|x\|\|y\|$, with equality if and only if x and y are linearly dependent.*

Proof. If $y = 0$, the assertion is clear. Otherwise put $t = \langle x, y\rangle / \|y\|^2$. Then

$$0 \le \|x - ty\|^2 = \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2}.$$

Rearranging proves the inequality. Equality holds precisely when $x = ty$, that is, when the vectors are linearly dependent. ■

Theorem 4.7.7 (Properties of the Norm)**.** *For $x, y \in \mathbb{K}^n$ and $a \in \mathbb{K}$:*

(i) $\|x\| \ge 0$, *and* $\|x\| = 0$ *exactly when* $x = 0$;

(ii) $\|ax\| = |a|\|x\|$;

(iii) $\|x + y\| \le \|x\| + \|y\|$ *(triangle inequality).*

Proof. The first two properties follow from the definition. For the third, Cauchy–Schwarz gives

$$\begin{aligned}\|x + y\|^2 &= \|x\|^2 + 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2 \\ &\le \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.\end{aligned}$$

Taking square roots completes the proof. ■

Example 4.7.8. Using $x = \begin{bmatrix} 1+i \\ 2 \\ -i \end{bmatrix}$ and $y = \begin{bmatrix} 2i \\ 1-i \\ 3 \end{bmatrix}$ from the previous example, verify the Cauchy–Schwarz Inequality $|\langle x, y\rangle| \le \|x\|\|y\|$.

Solution. From the previous calculations:

$$\langle x, y\rangle = 4 - 3i \implies |\langle x, y\rangle| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5.$$

$$\|x\| = \sqrt{7}, \quad \|y\| = \sqrt{15} \implies \|x\|\|y\| = \sqrt{7 \times 15} = \sqrt{105} \approx 10.246.$$

Comparing the values: $5 \le \sqrt{105} \implies |\langle x, y\rangle| \le \|x\|\|y\|$. This verifies the Cauchy–Schwarz Inequality. ■

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4.7.3 Pythagorean Theorem and Parallelogram Law

Theorem 4.7.9 (Pythagorean Theorem). *If $x \perp y$, then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.*

Proof. Expand $\langle x+y, x+y\rangle$ and use $\langle x, y\rangle = \langle y, x\rangle = 0$. ■

Theorem 4.7.10 (Parallelogram Law). *For $x, y \in \mathbb{K}^n$,*

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

Proof. Expand both squared norms using the inner product. Their mixed terms cancel upon addition. ■

Example 4.7.11. Let $x = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^2$. Verify the Parallelogram Law $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$.

Solution. Compute the norms of x and y: $\|x\|^2 = 3^2 + 1^2 = 10 \implies 2\|x\|^2 = 20$, $\|y\|^2 = 1^2 + 2^2 = 5 \implies 2\|y\|^2 = 10$. Thus, the right-hand side is $2\|x\|^2 + 2\|y\|^2 = 20 + 10 = 30$.

Now compute $x+y$ and $x-y$:

$$x+y = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \implies \|x+y\|^2 = 4^2 + 3^2 = 25,$$

$$x-y = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \implies \|x-y\|^2 = 2^2 + (-1)^2 = 5.$$

The left-hand side is $\|x+y\|^2 + \|x-y\|^2 = 25 + 5 = 30$.

Since $30 = 30$, the Parallelogram Law holds. ■

4.7.4 Orthonormal Bases and Fourier Expansion

Definition 4.7.12 (Orthonormal Basis). An **orthonormal basis** of $\mathbb{K}^n$ is a basis that is an orthonormal set.

Theorem 4.7.13 (Fourier Expansion). *If $\{u_1, \ldots, u_n\}$ is an orthonormal basis for $\mathbb{K}^n$, then*

$$x = \sum_{i=1}^{n} \langle x, u_i\rangle u_i.$$

Proof. Write $x = \sum_i c_i u_i$ and take the inner product with u_j in the second argument. Orthonormality gives $c_j = \langle x, u_j\rangle$. ■

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Theorem 4.7.14 (Parseval's Identity). *For such an orthonormal basis,*

$$\|x\|^2 = \sum_{i=1}^{n} |\langle x, u_i \rangle|^2.$$

Proof. Substitute the Fourier expansion into $\langle x, x \rangle$ and use $\langle u_i, u_j \rangle = \delta_{ij}$. ■

Example 4.7.15. Show that

$$\mathcal{U} = \left\{ u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

is an orthonormal basis for $\mathbb{R}^3$, and express $x = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$ using Fourier coefficients.

Solution. We check orthonormality by computing inner products $\langle u_i, u_j \rangle$:

$$\langle u_1, u_1 \rangle = \frac{1}{2}(1+1+0) = 1, \quad \langle u_2, u_2 \rangle = \frac{1}{2}(1+1+0) = 1, \quad \langle u_3, u_3 \rangle = 1.$$

$$\langle u_1, u_2 \rangle = \frac{1}{2}(1-1+0) = 0, \quad \langle u_1, u_3 \rangle = 0, \quad \langle u_2, u_3 \rangle = 0.$$

Thus, $\mathcal{U}$ is an orthonormal set of 3 vectors in $\mathbb{R}^3$, making it an orthonormal basis.

Now compute the Fourier coefficients $c_i = \langle u_i, x \rangle$:

$$c_1 = \langle u_1, x \rangle = \frac{1}{\sqrt{2}}(1(3) + 1(1) + 0(4)) = \frac{4}{\sqrt{2}} = 2\sqrt{2},$$
$$c_2 = \langle u_2, x \rangle = \frac{1}{\sqrt{2}}(1(3) - 1(1) + 0(4)) = \frac{2}{\sqrt{2}} = \sqrt{2},$$
$$c_3 = \langle u_3, x \rangle = 0(3) + 0(1) + 1(4) = 4.$$

Applying the Fourier expansion theorem: $x = 2\sqrt{2}u_1 + \sqrt{2}u_2 + 4u_3$. Verifying Parseval's identity: $\|x\|^2 = 3^2 + 1^2 + 4^2 = 26$, and $c_1^2 + c_2^2 + c_3^2 = (2\sqrt{2})^2 + (\sqrt{2})^2 + 4^2 = 8 + 2 + 16 = 26$. ■

4.7.5 Gram–Schmidt Orthogonalization

Theorem 4.7.16 (Gram–Schmidt Process). *Every linearly independent set $\{v_1, \ldots, v_k\}$ in $\mathbb{K}^n$ can be transformed into an orthonormal set $\{u_1, \ldots, u_k\}$ with the same successive spans. Set*

$$w_1 = v_1, \quad u_1 = \frac{w_1}{\|w_1\|}, \qquad w_m = v_m - \sum_{j=1}^{m-1} \langle v_m, u_j \rangle u_j, \quad u_m = \frac{w_m}{\|w_m\|} \quad (m \geq 2).$$

Proof. Each w_m is orthogonal to every preceding u_j. Linear independence ensures $w_m \neq 0$, so normalization is possible. The successive spans remain unchanged because only combinations of earlier vectors are subtracted. ■

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Example 4.7.17. Apply the Gram–Schmidt process to transform the linearly independent set $\{v_1, v_2\} \subset \mathbb{R}^3$ into an orthonormal set $\{u_1, u_2\}$, where:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T.$$

Solution. Construct the first orthogonal vector $w_1 = v_1$:

$$w_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \implies \|w_1\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}.$$

Normalize w_1 to get u_1:

$$u_1 = \frac{w_1}{\|w_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Construct the second orthogonal vector $w_2 = v_2 - \langle u_1, v_2 \rangle u_1$: First, compute the inner product:

$$\langle u_1, v_2 \rangle = \frac{1}{\sqrt{2}}(1(1) + 1(0) + 0(1)) = \frac{1}{\sqrt{2}}.$$

Then compute w_2:

$$w_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}.$$

Compute the norm of w_2:

$$\|w_2\| = \sqrt{(1/2)^2 + (-1/2)^2 + 1^2} = \sqrt{1/4 + 1/4 + 1} = \sqrt{3/2} = \frac{\sqrt{6}}{2}.$$

Normalize w_2 to get u_2:

$$u_2 = \frac{w_2}{\|w_2\|} = \frac{2}{\sqrt{6}} \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}.$$

The resulting orthonormal set is $\{u_1, u_2\}$. ■

4.7.6 Distance and Orthogonal Projection

For $x, y \in \mathbb{K}^n$, distance is $d(x, y) = \|x - y\|$. The projection of x onto a non-zero vector u is

$$\text{proj}_u(x) = \frac{\langle x, u \rangle}{\|u\|^2} u.$$

The difference $x - \text{proj}_u(x)$ is orthogonal to u.

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Example 4.7.18. Let $x = \begin{bmatrix}1\\2\\2\end{bmatrix}$ and $y = \begin{bmatrix}0\\3\\4\end{bmatrix}$ in $\mathbb{R}^3$. Calculate the distance $d(x,y)$, the angle θ between x and y, and the orthogonal projection $\text{proj}_y(x)$.

Solution. First, compute the individual norms and inner product: $\|x\| = \sqrt{1^2+2^2+2^2} = \sqrt{9} = 3$, $\|y\| = \sqrt{0^2+3^2+4^2} = \sqrt{25} = 5$, $\langle x,y\rangle = 1(0)+2(3)+2(4) = 0+6+8 = 14$.

To compute the distance $d(x,y) = \|x-y\|$:

$$x - y = \begin{bmatrix}1\\-1\\-2\end{bmatrix} \implies d(x,y) = \sqrt{1^2+(-1)^2+(-2)^2} = \sqrt{1+1+4} = \sqrt{6}.$$

To compute the angle θ:

$$\cos\theta = \frac{\langle x,y\rangle}{\|x\|\|y\|} = \frac{14}{3\times 5} = \frac{14}{15} \implies \theta = \arccos\left(\frac{14}{15}\right) \approx 0.367 \text{ radians } (21.04^\circ).$$

To compute the orthogonal projection $\text{proj}_y(x)$:

$$\text{proj}_y(x) = \frac{\langle y,x\rangle}{\|y\|^2}y = \frac{14}{25}\begin{bmatrix}0\\3\\4\end{bmatrix} = \begin{bmatrix}0\\42/25\\56/25\end{bmatrix}.$$

■

Exercise 4.7.19. Solve the following exercises.

1. Let $A = \begin{bmatrix}2 & 1\\0 & 3\end{bmatrix} \in M_2(\mathbb{R})$ and $p(x) = x^3 - 3x^2 + 2x - 1$. Evaluate the matrix polynomial $p(A)$.
2. Let $A = \begin{bmatrix}0 & 1 & 0\\0 & 0 & 2\\0 & 0 & 0\end{bmatrix} \in M_3(\mathbb{R})$. Show that A is nilpotent, state its index of nilpotency k, and evaluate $(I_3 - A)^{-1}$ and e^A as finite matrix polynomials.
3. Prove that if $A \in M_n(\mathbb{K})$ is an idempotent matrix ($A^2 = A$), then every eigenvalue of A is either 0 or 1.
4. Find the characteristic polynomial $p_A(\lambda)$ and the minimal polynomial $m_A(x)$ for the matrix:
$$A = \begin{bmatrix}3 & 1 & 0\\0 & 3 & 0\\0 & 0 & 2\end{bmatrix}.$$
5. Compute the characteristic polynomial $p_A(x)$ and minimal polynomial $m_A(x)$ for:
$$A = \begin{bmatrix}1 & 0 & 0\\2 & 1 & 0\\0 & 0 & 1\end{bmatrix}.$$

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6. Let $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{bmatrix}$.
   (a) Find the characteristic polynomial of A.
   (b) List the possible minimal polynomials.
   (c) Determine the minimal polynomial by testing the candidates directly.

7. Find the eigenvalues, algebraic multiplicities, geometric multiplicities, and a basis for each eigenspace of:
$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

8. Let $A, B \in M_n(\mathbb{K})$ be similar matrices ($B = P^{-1}AP$). Prove that A and B have the same characteristic polynomial, trace, determinant, and eigenvalues.

9. Prove that if $v_1, v_2, \ldots, v_k \in \mathbb{K}^n$ are eigenvectors of a matrix A corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$, then $\{v_1, v_2, \ldots, v_k\}$ is linearly independent.

10. Let $A \in M_n(\mathbb{K})$ be an invertible matrix. If λ is an eigenvalue of A with corresponding eigenvector x, prove that $\lambda \neq 0$ and that λ^{-1} is an eigenvalue of A^{-1} with the same eigenvector x.

11. Verify the Cayley–Hamilton Theorem for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Use the theorem to compute A^{-1} and A^4.

12. Verify the Cayley–Hamilton Theorem for the matrix:
$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix},$$
and use it to find A^{-1}.

13. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Using the Cayley–Hamilton Theorem, evaluate the high-degree matrix polynomial: $f(A) = A^5 - 2A^4 - 3A^3 + 2A^2 - 3A + 5I_2$.

14. Given $x = \begin{bmatrix} 2+i \\ 3 \\ -2i \end{bmatrix}$ and $y = \begin{bmatrix} 1-i \\ i \\ 2 \end{bmatrix}$ in $\mathbb{C}^3$:
   (a) Compute the complex inner product $\langle x, y \rangle = x^* y$.
   (b) Compute the norms $\|x\|$ and $\|y\|$.
   (c) Verify the Cauchy–Schwarz Inequality $|\langle x, y \rangle| \leq \|x\| \|y\|$.

15. Prove the Parallelogram Law for vectors in $\mathbb{K}^n$:
$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

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16. Prove the Pythagorean Theorem in $\mathbb{K}^n$: if $x, y \in \mathbb{K}^n$ are orthogonal ($x \perp y$), then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.

17. Show that $S = \left\{ u_1 = \frac{1}{3}\begin{bmatrix}2\\2\\1\end{bmatrix}, u_2 = \frac{1}{3}\begin{bmatrix}-2\\1\\2\end{bmatrix}, u_3 = \frac{1}{3}\begin{bmatrix}1\\-2\\2\end{bmatrix} \right\}$ is an orthonormal basis for $\mathbb{R}^3$. Find the Fourier coefficients of $x = \begin{bmatrix}3\\0\\3\end{bmatrix}$ relative to S.

18. Apply the Gram–Schmidt process to transform the linearly independent set $\{v_1, v_2, v_3\} \subset \mathbb{R}^3$ into an orthonormal set $\{u_1, u_2, u_3\}$, where:
$$v_1 = \begin{bmatrix}1\\1\\0\end{bmatrix}, \quad v_2 = \begin{bmatrix}1\\0\\1\end{bmatrix}, \quad v_3 = \begin{bmatrix}0\\1\\1\end{bmatrix}.$$

19. For the vectors $x = \begin{bmatrix}2\\1\\2\end{bmatrix}$ and $y = \begin{bmatrix}1\\2\\2\end{bmatrix}$ in $\mathbb{R}^3$:
    (a) Compute the distance $d(x,y) = \|x-y\|$.
    (b) Find the angle θ between x and y.
    (c) Compute the orthogonal projection $\text{proj}_y(x)$.

20. Prove Parseval's Identity: if $\{u_1, u_2, \ldots, u_n\}$ is an orthonormal basis for $\mathbb{K}^n$, then for any vector $x \in \mathbb{K}^n$:
$$\|x\|^2 = \sum_{i=1}^{n} |\langle u_i, x\rangle|^2.$$

Concluding Note to Students

Congratulations!

"Mathematics is not about numbers, equations, or computations: it is about understanding."

— William Paul Thurston

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DEAR STUDENTS,

Congratulations on successfully completing the course on **Theory of Matrices (MMT126J)**! Reaching this milestone reflects your dedication, perseverance, and growing mathematical maturity.

Throughout this book, you have mastered the essential tools of linear algebra—transitioning from matrix algebra, structural symmetries, and elementary row operations (Units I–II) to linear independence, basis, rank, trace, spectral theory, the Cayley–Hamilton theorem, and inner product spaces (Units III–IV).

Matrix theory is the universal language of modern science, technology, engineering, and advanced mathematics. The conceptual clarity, abstract reasoning, and analytical problem-solving skills you have cultivated here will serve as an enduring foundation for your future studies and competitive examinations such as **CSIR NET-JRF, GATE, SET, IIT-JAM, and Ph.D. entrance tests**.

Keep exploring, stay curious, and carry this logical confidence forward into every domain of your academic journey. I wish you every success and fulfillment in all your future endeavors!

DR. AIJAZ
Assistant Professor of Mathematics
JK Higher Education Department
Website: `aijaz.vercel.app`

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